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Linear Combination of Columns

Rewriting the following system of equations into column form we get,

$$2x - y = 0$$

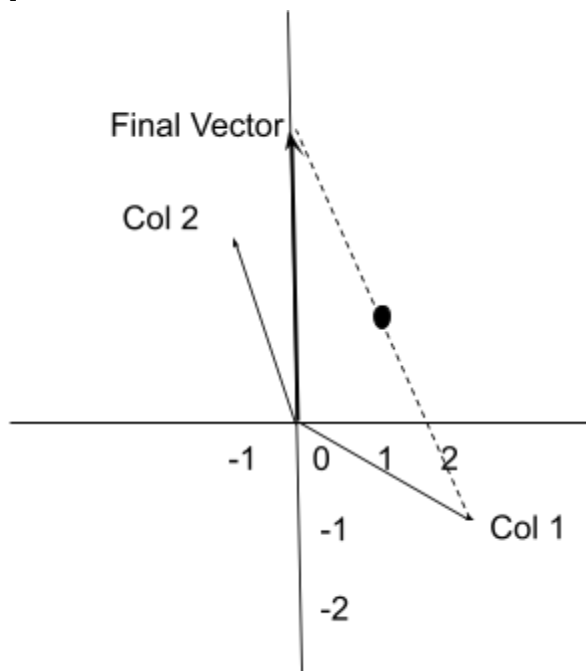
$$-x + 2y = 3$$

This column form is also called the Linear Combination of Columns.

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

The two vectors on LHS can be considered as two columns and we have to substitute values for x and y to combine the two “columns” and get the final vector which is on the RHS.

Say we take $x = 1$ and $y = 2$ (this is a right combination because we know that $(1, 2)$ satisfies both equations), then we can see this linear combination graphically to give us the result vector $[0 \ 3]$.



So we added two units of Col 2 to Col 1 to reach point $(0, 3)$, i.e. vector $[0 \ 3]$ which is the result vector.

Notation from Prof. Strang:

If you plot the equations directly in a graph, then that's called the “row picture”

If you plot the equations via the linear combination of columns form, like the image above, then that's called the “column picture”.

And finally there's the matrix form, $AX = B$ (this is formal terminology and not just notation).

Row vs Column Picture

This was a 2-dimensional problem. What if it was a 3-dimensional problem, i.e. 3 equations and 3 unknowns?

Then when you plot out the equations, you will get a plane. Each plane would contain all the points that will satisfy the given equations. Any two of these planes will intersect with each other at some line (in 2D, there are lines that intersect at points, so you can see how it works) and then all 3 of these planes will intersect at a point and it will represent the solution. Ofcourse this only works if the system of equations actually has a solution, i.e. no parallel planes, stuff like that.

Plotting these planes out becomes really annoying as the system grows more complex, so it is better to use column picture notation and draw the equations as column vectors on the graph.

Using Linear Combination to fill 3D space

Using this column picture a bit forward, ask the question, is it possible to find the unknowns, i.e. x , y and z for every possible value of b ?

Rephrasing this question, is it possible to fill the entire 3D space using linear combinations of the 3 vectors associated with x , y and z ?

The answer is that it depends on the column vectors themselves.

If all 3 vectors lie in the same plane, then all of their linear combinations would also lie in the same plane and so they wouldn't even be able to break out of their plane, let alone cover the entire 3D space. This would happen when the matrix is non-invertible or singular.

Multiplying two Matrices

You know the normal way of doing this each entry at a time, but Prof. Strang does this using columns obviously ;)

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

So each element of the column vector is multiplied as a scalar to the corresponding column of the matrix, i.e. the first element of the vector is multiplied with the first column, the second element of the vector is multiplied with the second column and so on and they are added up.

Remember that AX is a linear combination of the columns of A as it will make calculations and intuition much easier with more complex matrices.

Extra Note:

What if you are multiplying with a row vector instead of a column vector? This will be helpful in case you are doing row operations which is what the Elimination Method is all about.

Example,

$$\begin{bmatrix} 1 & 2 & 7 \end{bmatrix} \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} = \text{1x3 row vector} \\ = [1 \cdot \text{row1} \quad 2 \cdot \text{row2} \quad 7 \cdot \text{row3}]$$

So this is the row-wise way of looking at matrix multiplication and also tells you that matrix multiplication is a linear combination of rows.

Note: These ways of looking at matrix multiplication are **applicable for matrices as well** and **not just restricted to multiplications where one is a vector**. You just have to repeat the steps for each row or column depending on whether you are doing the row-wise view or column-wise view.

Remember that if you are going to be performing a multiplication and want to use the **row-wise picture**, then it would have to be **multiplied on the LHS of the matrix** and if you are going to use the **column-wise picture**, then it would have to be **multiplied on the RHS of the matrix**. Don't worry if this doesn't make sense because after the Elimination Method section, it will make sense.

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Elimination Method

This process basically involves converting a given matrix ("A" matrix from $AX=B$ form) into an upper triangular matrix.

Key points,

- In each step of this process of conversion, we must keep one element as our pivot, based on which, the elements from the next row are made zero.
- These pivot elements are the elements from the primary diagonal of the matrix.
- The operations that are performed on the "A" matrix would also have to be performed on the "B" vector on the RHS.
- If any of the pivot elements are zero then we cannot proceed further with the elimination process.

Success with Elimination

Example,

$$\begin{aligned}x + 2y + z &= 2 \\3x + 8y + z &= 12 \\4y + z &= 2\end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \Rightarrow A_{11} = \text{Pivot} \Rightarrow R_2 \rightarrow R_2 - 3R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \Rightarrow A_{22} = \text{Pivot} \Rightarrow R_3 \rightarrow R_3 - 2R_2 \Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

Now we have reduced A to an upper triangular matrix, call it U.

Note:

- If A_{31} was not zero, then we would have had to perform an extra operation for that element as well.
- The determinant of an upper triangular matrix is just the product of its diagonal elements.

Failure with Elimination

How could this process have failed?

- If any of the pivot elements were zero.
- The pivots can be zero right from the start or they can become zero as we perform the operations, in which case we must do something to make it non-zero.
- One of these somethings is exchanging that row for a lower row so that the required pivot element is no longer zero, for example, in the above example, if A_{22} was zero, then we could have exchanged R_2 and R_3 to get 4 up at A_{22} .

This method of exchanging for a lower row to fix the pivot works if there are any lower rows that have non-zero values in the pivot position. But what if it's not? What if it's the last row?

In that case, you are stuck and can do no more operations.

This also means that this matrix is non-invertible, i.e. singular, i.e. determinant = 0.

Back-Substitution

Now whatever was done on the LHS of the equation, i.e. matrix A, also has to be done on the RHS, i.e. vector B.

To do this, we “augment” the matrix A by adding vector B as the last column. This augmented matrix will go through the same process as was done in the solved example and this ensures that the RHS vector also is modified appropriately. MATLAB actually solves “A” matrix first and only then solves “B”, but for manual calculations it makes more sense to do them both simultaneously.

Also, call the new column that is obtained after performing calculations, C. So A becomes U and B becomes C.

Example, This is what happens to the augmented matrix of A and B from the above example.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{bmatrix}$$

What is back-substitution then?

Write the equations from the augmented matrix.

$$x + 2y + z = 2$$

$$2y - 2z = 6$$

$$5z = -10$$

Now we can clearly start solving these equations and we get the following solutions by solving for z first and then move backwards to solve y and finally x.

$$z = -10/5 = -2$$

$$y = \frac{6 + 2(-2)}{2} = 1$$

$$x = 2 - 2(1) - (-2) = 2$$

And this is the process of back-substitution.

Representing Matrix Operations

In the solved examples, we represented the matrix operations conveniently using notations like, $R_2 \rightarrow R_2 - 3R_1$

But in the actual working we need to represent each operation as a matrix multiplication operation, where we determine the matrix that we need to multiply with to get the correct output. The row-wise picture of performing Matrix Multiplication will help a lot here.

We need to perform the operation, $R_2 - 3R_1$.

So our picture would look like the following,

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

Now we can use the row-wise picture of Matrix Multiplication to fill up the ? matrix.

Let's note this equation as, $CA=B$

We only want 1*row3 of A in B, so we put row 3 of C as $[0 \ 0 \ 1]$.

We only want 1*row1 of A in B, so we put row 1 of C as $[1 \ 0 \ 0]$.

Now in row 2 of B, we want, $-3 * \text{row1}$ and $1 * \text{row2}$, so we put row 2 of C as $[-3 \ 1 \ 0]$.

This gives us the final picture as,

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

Let's call the C matrix as E instead (E for Elimination) and denote it as E_{21} because this is the matrix that we used to fix the (2, 1) position.

Now we have to do the next operation in matrix language, this was,
 $R_3 \rightarrow R_3 - 2R_2$

After working out the calculations like before, we get E_{32} as follows,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

Now we can represent the entire sequence of Elimination in a single Matrix operation as follows,
 $E_{32} \cdot (E_{21} \cdot A) = U$

Now can we represent the two E matrices in a single matrix?

Yes. If we move the parentheses around and multiply the two E matrices first.

This is valid because although matrix multiplication is not commutative and sequence is important, we can adjust the parentheses without it affecting the final answer, i.e. **matrix multiplication is associative**.

$$(E_{32} \cdot E_{21}) \cdot A = U$$

Detour: Permutation Matrix

Elementary Matrix

An *elementary matrix* is a matrix which differs from the identity matrix by one single elementary row operation

The Permutation Matrix is an Elementary matrix which exchanges two rows.

For example,

$$\begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

Using the row picture discussed earlier you can easily figure out that the Permutation Matrix in this case is,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Point to remember which will come up later as well:

Row Operation: Matrix Multiplication on the Left

Column Operation: Matrix Multiplication on the Right

Intro to Inverse Matrices

This was the operation that we had,

$$(E_{32} \cdot E_{21}) \cdot A = U$$

And so we can multiply the two E matrices to get a matrix that will encompass all the operations that we did on A to get U.

But this is cumbersome and we want another way.

A potential way is instead of going from A to U, we can come from U to A.

So this means that we want matrices that undo operations that were performed on A.

The inverse matrix of each E matrix represents the undo operation matrix for it so to speak.

This is because if you multiply the inverse matrix E^{-1} with the matrix E, then we get the Identity Matrix I, which represents no operation basically in the world of matrices.

All matrices discussed till now are invertible so we won't run into that case yet and we can quickly get the inverse of the E matrices using the row picture as follows,

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

E_{21} matrix just subtracted 3 times row 1 from row 2, so we can undo that by adding 3 times row 1 back to row 2 which is how we get the inverse as,

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This was easy to compute because it was easy to picture what operations E_{21} matrix was doing which will definitely not be the case every time.

Multiplication and Inverse Matrices

Ways of Multiplying Matrices

Method 1: Element-wise (Standard matrix multiplication)

Method 2: Expressing multiplication as linear combination of columns (of the matrix on the left of multiplication)

Method 3: Expressing multiplication as linear combination of rows (of the matrix on the right of multiplication)

These 3 methods are already discussed in the Matrix Multiplication section.

Method 4: Expressing multiplication as sum of product of rows and columns.

$AB = \text{Sum}(\text{Columns of } A * \text{Rows of } B)$

Example,

$$\begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}$$

Method 5: Multiplying matrices by blocks

This is not a separate method really but an optimization that can be used to multiply large matrices and then you can use any one of the above methods to solve it completely.

You can chop up both matrices in the multiplication into equal sized blocks which are themselves matrices and then the blocks can participate in matrix multiplication as if they were matrix elements,

Consider the following illustration, multiplication $A * B$ where both are square and of the same order for simplicity's sake and they are divided into 4 blocks A_1, A_2, A_3, A_4 and B_1 to B_4 .

Now the final multiplication will look as follows,

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1 \cdot B_1 + A_2 \cdot B_3 & A_1 \cdot B_2 + A_2 \cdot B_4 \\ A_3 \cdot B_1 + A_4 \cdot B_3 & A_3 \cdot B_2 + A_4 \cdot B_4 \end{bmatrix}$$

Inverses

$$A^{-1} \cdot A = I = A \cdot A^{-1}$$

When the inverse is on the left, it's called the left inverse and right inverse when it's on the right.

The inverse is valid only when it actually exists, i.e. the matrix A is invertible.

For square matrices that are invertible, it is guaranteed that the left inverse and right inverse are the same. And for rectangular matrices, this isn't the case, they may either have the left inverse or the right inverse or neither of them and for this reason they are never invertible.

Non-invertible matrices = Singular matrices = Determinant is zero

Intuition behind why Singular Matrices are Non-invertible

$$\text{Let } A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

A is singular because its determinant is zero and therefore has no inverse.

But why?

Imagine that if the inverse exists and it can be multiplied with A to get the matrix I.

Then that means that we have to get the identity matrix as a linear combination of columns of matrix A.

But this is not possible. Because both of the columns of A lie on the same line (imagine the column picture of matrix A, they are both on the same line), so they can never be made to point to another line or point irrespective of what they're multiplied or divided by.

Another more prominent reason for why this is not possible is because you can find a vector X such that, $AX = 0$

Note: X cannot be vector of just zeroes because that's cheating :P

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now how does this prove that A is non-invertible?

Assume that A^{-1} exists and try multiplying it with the $AX = 0$ equation.

We will get $X = 0$, but that is not the case because we just found that there is a valid non-zero vector X which satisfies the equation. This means that our assumption A^{-1} exists is false.

Gauss-Jordan Method

(solve two equations at once)

Consider the example,

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now using the column method, we can break it into two systems of equations as follows,

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

&

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Now the Gauss-Jordan method combines the two systems to form the augmented matrix AI (I =Identity Matrix that's on RHS), then converts A to the identity matrix via elimination method and finally when A does become the same as I , whatever the B part of the augmented matrix is, that is the inverse of A .

[

A simple way I remember this as is,

$$A * A^{-1} = I$$

If you convert to A to I , so that the new equation is, $I * A^{-1} = C$, then $A^{-1} = C$ and that is what this method does.

]

$$\text{Augmented Matrix} = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix}$$

At this point we have an upper triangular matrix which is where the Gauss method stops but this epic combined approach goes further to convert it into an identity matrix.

$$R_1 \rightarrow R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{bmatrix}$$

Now we can say that,

$$A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$$

Why is this the case?

We know that each elimination operation can be boiled down to an elimination matrix and the product of all these elimination matrices gives us a big Elimination Matrix, E .

If we multiply $E * A$ directly, then we'll get the output of the elimination process which in this case is I . But if we multiply two matrices and are getting the identity matrix, then this means that that is the inverse, E is actually the inverse then.

And how did this E come to be on the right half of the augmented matrix?

Remember that the Gauss-Jordan elimination process is performed on the augmented matrix $[A \ I]$

So,

$$E \cdot [A \ I] = [E \cdot A \ E] = [I \ E]$$

And this is how E is on the right half of the augmented matrix which is actually the inverse.

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Inverse of Product of Matrices

$$(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$$

Provided A and B are invertible.

Why?

$$\text{We know that, } (AB) \cdot (AB)^{-1} = I$$

$$\text{Let } (AB)^{-1} = B^{-1}A^{-1}, \text{ then}$$

$$= A(B \cdot B^{-1})A^{-1} \text{ (By associative property)}$$

$$= A \cdot I \cdot A^{-1}$$

$$= A \cdot A^{-1}$$

$$= I$$

$$\text{Similarly we can prove that, } (B^{-1}A^{-1})^{-1} = AB$$

Inverse of Transpose

$$A \cdot A^{-1} = I$$

Perform Transpose on both sides & we get,

$$(A^{-1})^T \cdot A^T = I$$

Note : I remains the same upon transpose.

This tells us that the inverse of A^T is $(A^{-1})^T$

Factorization of $A = LU$

Through the process of elimination we reduced A to an upper triangular matrix U using just row operations.

We combined all the E matrices associated with each elimination operation via their product into a single matrix called E.

So our equation is,

$$E \cdot A = U$$

Example to explain this better,

$$\begin{aligned} E \cdot A &= \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = U \end{aligned}$$

$$A = LU$$

Find L using values of A & U with the row multiplication logic,

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

$$\text{Therefore, } L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

You will find that $L = E^{-1}$

This is because if you move E to the RHS in the equation $E \cdot A = U$, then,

$$A = E^{-1} \cdot U = L \cdot U$$

L = Lower Triangular Matrix!

Why do we prefer matrix L over E?

This can be illustrated with the help of a 3x3 matrix example of how calculations become simpler when it's L instead of E. That is, instead of multiplying all the E matrices, move them to the RHS so that they become inverses (also their order gets inverted) and then multiply them to get L. Look at how simple matrix L is as compared to matrix E.

Basically, if there's no row exchanges allowed then the multipliers go directly into L whereas they can recombine into various forms while forming E.

How many operations does elimination on a matrix of order n require?

An operation doesn't just mean a row operation, we are talking about multiplications and subtractions are performed as part of a row operation.

Prof. Strang assumed that one operation consists of a Multiplication and a Subtraction (can be an addition too in case the matrix elements are negative).

Now consider $n = 100$, so a 100x100 matrix.

We have to first fix the (2,1) element, for this we subtract from row 1, so that (2,1) element becomes 0. So there are essentially 100 operations occurring because we have to perform the operation for the entire row and number of elements in a row = 100.

Now we have to fix the (3,1) element, this also takes 100 operations.

And we repeat this for 97 more rows.

In total if we want to fix the first column, we have to perform 99×100 operations

That is, $(n - 1) \times n$ operations.

Since we are dealing with measuring complexities, just assume it to be n^2 .

So fixing the first column takes about 100^2 operations.

For fixing the second column, we don't have to fix the first two rows and the first column is not involved too, so the total number of operations becomes, 98×99 operations.

Approximate this to be 99^2 .

Now we can continue this pattern, for the third column it will be 98^2 and so on.

$$\begin{aligned}\text{Total number of operations} &\approx n^2 + (n - 1)^2 + \dots + 2^2 + 1^2 \\ &\approx \frac{n^3}{3}\end{aligned}$$

However, this was only for matrix A which is on the left side, **for column vector B on the right side, we will have to do approximately n^2 calculations.**

Permutation Matrices

If we allow row exchanges in the matrix operations, then as discussed before, we'll require permutation matrices which are just rearrangements of the corresponding identity matrix (i.e. same order identity matrix as that of A).

Consider the 3x3 matrix, then what are the permutation matrices,

We can exchange zero rows = Identity Matrix, exchange 2 rows, i.e. 1 row exchanged with the other, exchange 3 rows, i.e. all 3 rows are shifted from their position.

And these are the permutation matrices,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Two row exchange,

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Three row exchange,

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Thus, there are a total of 6 permutation matrices for 3x3 matrices.

These 6 P matrices form their own group over the matrix multiplication operation (group => Discrete Math concept).

Because, if you multiply any two of the matrices with each other, the result would be a matrix from these 6 only (closed, i.e. algebraic structure).

Matrix multiplication is associative (i.e. semi-group)

The identity matrix would be the identity element for this group (i.e. monoid)

If you inverse any of the matrices, then you will get one of these 6 matrices only. (i.e. group)

If you look at the inverses of the matrices, you will find that the inverses of the two row exchange matrices are the same.

And for each three row exchange matrix, the inverse is the other three row exchange matrix.

This is happening because for this group,

$$P^{-1} = P^T$$

How many permutation matrices exist for order 4?

This would be 24 matrices.

Why?

A permutation matrix can be obtained by placing 1s in an nxn matrix such that no two 1s are on the same column or row. Think of the n-Queens problem or a Sudoku puzzle for an analogy.

So we can use counting to get the right answer here,

In a 4x4 matrix, traverse row-wise to place a 1 in any column.

At row 1, we can place a 1 in any of the 4 columns, so 4 choices.

At row 2, we can place a 1 in the remaining 3 columns, so 3 choices.

For row 3, its 2 choices and for row 4, its 1 choice.

So in total we have $4 * 3 * 2 * 1 = 4! = 24$ outcomes = 24 permutation matrices.

Therefore for a matrix of order n,

Number of permutation matrices = $n!$

The equation $A = L \cdot U$, is for the case when the elimination process needed no row exchanges.

However when row exchanges come into the picture, we require permutation matrices which gives us the new equation,

$$P \cdot A = L \cdot U$$

When no row exchanges required, then, $P = I$

This matrix P will ensure that P.A is a "good matrix", i.e. no zeroes in the pivot position and elimination is possible.

Symmetric Matrices & Transpose

$$A^T = A$$

This is only possible if A is a square matrix.

However for any matrix A (square or rectangular),

$$A \cdot A^T = B$$

where, B = Symmetric

You can try this formula with an example and you'll notice the matrix elements that are being multiplied are the same for the symmetric positions (i.e. same for B_{ij} & B_{ji}), just the order of the numbers are reversed in the multiplications.

But why do all that crap?

If you want to check any matrix is symmetric, just take its transpose.

Here we want to check whether $A \cdot A^T$ is symmetric, so take its transpose,

[

And remember the rule for transpose of product of matrices, which is the same as that of inverses,

$$(A \cdot B)^T = B^T \cdot A^T$$

]

$$\begin{aligned} (A \cdot A^T)^T &= (A^T)^T \cdot A^T \\ &= A \cdot A^T \end{aligned}$$

The result is the same as the input proving that the matrix is symmetric.

Vector Spaces

A vector space refers to a bunch of vectors that allow us to perform vector operations, such as addition, multiplication by a scalar and linear combinations.

Definition I found online: A space consisting of vectors, together with the associative and commutative operation of addition of vectors and the associative and distributive operation of multiplication of vectors by scalars.

The vector space R^2 , refers to the two real numbers, i.e. it encompasses all 2-dimensional real number vectors such as, $[0\ 0]$, $[3\ 2]$, $[\pi\ e]$, etc.

What makes a vector space not a vector space?

Say if you remove any point from R^2 , then it is possible that some vectors can undergo an operation (addition or multiplication) to get its value equivalent to that point which has been removed from the vector space. In this case, the answer is not valid because for that vector space, the point doesn't exist and so the operation is invalid which makes the vector space not a vector space.

For example, remove the origin from R^2 and now any operation that gives $[0\ 0]$ as the answer will make the resultant vector space invalid.

Basically what I understood from this is that a vector space has to obey the closure property over the multiplication (scalar and vector) and addition operations (and also linear combinations because it's just a combination of additions and scalar multiplications).

A space is a vector space if,

$$v + w \in \text{space}, \text{ where } v, w \in \text{space}$$

$$c \cdot v \in \text{space}, \text{ where } v \in \text{space} \ \& \ c \text{ is a scalar}$$

$$c \cdot v + d \cdot w \in \text{space}, \text{ where } v, w \in \text{space} \ \& \ c, d \in \text{scalars}$$

Rule 3 is just a consequence of Rules 1 & 2.

Note:

Assume that vectors refer to column vectors throughout this discussion.

Just because a component of a vector is zero doesn't mean it has one less dimension.

For example, $[1\ 2\ 0]$ is still a 3D vector despite it only having 2 non-zero components.

R^n refers to all vectors consisting of n real components (dimensions).

Subspaces

Vector spaces within another vector space.

Are there any subspaces within R^2 ?

Yes ofcourse.

Say you take the vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ within this potential subspace.

Now all scalar multiplications of this vector have to be present in the subspace, so you have to have $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\begin{bmatrix} -1 \\ -2 \end{bmatrix}$... all of these vectors in the subspace.

If you plot them out or visualize it, it's clear that all these vectors lie on the same line and we will essentially have to include the entire line that the vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ falls on to make scalar multiplications a closed operation.

Now we have to make sure that vector additions are closed in the subspace, this is already true because adding any two vectors on the same line will give you a vector that's also on the same line and so vector additions are closed.

So this subspace is finally valid and is a vector space as well.

Are all such lines valid subspaces?

No.

Because any such line has to pass through the origin to generate the zero vector. If it doesn't, then we multiply any vector on the line by 0 (scalar multiplication) to break the closure property.

Every subspace has to contain the zero vector.

Subspaces of $\mathbb{R}^2$

$\mathbb{R}^2$ itself.

All lines that pass through zero vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, L.

The zero vector alone, Z.

Using this same logic we can get the subspaces for $\mathbb{R}^3$,
 $\mathbb{R}^3$

All planes that contain the origin

All lines that pass through origin

Z

Subspaces using a Matrix: Column Space

$$\text{Let } A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$$

And now we want to make the columns of A, a subspace of $\mathbb{R}^3$.

We know that for this we need to add the zero vector, the addition of the two columns, all scalar multiples of both columns and **any linear combination of the two columns**.

We come to the conclusion that a space that contains all linear combinations of the two columns is a valid subspace of $\mathbb{R}^3$ and this is called the column space, denoted here by $C(A)$.

For a matrix A, $m \times n$, $C(A)$ is a subspace of $\mathbb{R}^m$.

In this specific case, it turns out that subspace $C(A)$ is a plane within R^3 . Most of the time it will be a plane but when columns are on the same line then $C(A)$ would also be a line.

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Combining Subspaces

Assume that we're in R^3 and P & L are some plane and line subspaces of it, respectively.

Then,

Is $P \cup L$, a subspace?

No!

This is because the addition of even any two vectors from P and L (one from each) would result in a vector that is not in the space.

Is $P \cap L$, a subspace?

Yes!

It is better to do this explanation with an example.

Say you take the intersection and are now about to do the linear combination,

$$c \cdot v + d \cdot w$$

Since vectors v, w belong to $P \cap L$, it means they both belong to P and L both, this also means that any scalar multiple of v and w also belongs to both P and L and this then means that any linear combination of v and w also belongs to both P and L and therefore it would also belong to $P \cap L$.

Column Spaces & $AX=B$

We saw before that the column space of a matrix A , $C(A)$, consists of all linear combinations of columns of A .

It was also concluded that this will not cover the whole n -dimensional space (where n =number of columns in A).

Now we apply the same analogy to $AX=B$.

Say we have a matrix A of order 4×3 (so column space is 4-dimensional) as,

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

Therefore can we find X for every possible value of B , i.e.,

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

This is asking the same question. Can we fill the entire 4-dimensional space using the linear combination of columns of A, i.e. column space of A?

This is how these two domains are connected.

And the answer to both of those questions is No!

You can also look at this way, how can you solve 4 equations with only 3 unknowns?

That should make it obvious.

What values of B can we solve X for??

[Assume that A is the same as shown before][

Clearly if B is zero vector then X can also be zero vector and we get an answer.

And that is always a given, if B = 0, then X = 0 and we get a solution.

$$\text{if } B = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \text{ then } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

(remember that for multiplication of X on RHS of A we can use the column picture, i.e. the linear combination of columns logic to quickly get the value for X).

Similarly we can find X for B where B = one of the columns of A.

We can now come to the realization that,

For some values of B you can actually solve 4 equations with just 3 unknowns and these values of B would fall under the column space of A.

This can be restated as, we can solve $AX = B$, when B is a linear combination of the columns of A, i.e. $C(A)$.

Independence of Columns

For the same value of A,

Are any of the columns independent?

A better way to ask that question is, can you throw away any columns from A and still retain the same column space?

The answer is yes, column 3 can be thrown away as it's just the addition of columns 1 & 2. So anything that column 3 can generate, we can generate via linear combinations of columns 1 & 2. Such base columns will be called as pivot columns, columns 1 & 2 in this case.

Could we have thrown away column 1 and retained column 3?

Yes.

Because if you keep both, then column 3 is dependent on 1 & 2 and column 1 is dependent on 2 & 3. If you remove one of the columns, then the dependency is gone from the matrix.

Nullspace of A

It contains all the X's for the equation $AX = 0$ and denoted by $N(A)$.

That is all X vectors that make $B = 0$.

For the matrix A described before, X is a 3D vector.

And therefore, the $N(A)$ is a subspace of $\mathbb{R}^3$.

For matrix A, $m \times n$, $N(A)$ is a subspace of $\mathbb{R}^n$.

Example,

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

We can easily find solutions for this small example and eventually get $N(A)$,

$$X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \dots$$

If you can figure it out then you realize that all values of X will be a multiple of,

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, i.e., C \cdot \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

This automatically includes the zero vector and we can take the value of c as negative so all those possibilities are covered as well.

If you visualize what is the space of vectors that you get, then it's a line in $\mathbb{R}^3$ that passes through the origin and $[1 \ 1 \ -1]$ and it's pretty clear that this is the Nullspace for A.

Prove that the Nullspace is actually a Vector Space

For this we have to prove that the space which consists of all vectors that solve $AX = 0$, satisfies the two conditions for being a vector space.

We can rephrase this pretty ingeniously, as, let $Av = 0$ and $Aw = 0$ where v and w are two valid solutions, then show that $v + w$ is also a solution to this, i.e. $A(v + w) = 0$.

Matrix multiplication follows distributive law, so this proof is pretty obvious,

$$\begin{aligned} A(v + w) &= A \cdot v + A \cdot w \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

We can also prove that any linear combination of these are also valid solutions,

$$\begin{aligned} A(c \cdot v + d \cdot w) &= A(c \cdot v) + A(d \cdot w) \\ &= c \cdot (A \cdot v) + d \cdot (A \cdot w) \text{ (scalars can be moved without violation)} \\ &= c \cdot 0 + d \cdot 0 \\ &= 0 \end{aligned}$$

If B wasn't the zero vector, then would X ever be a vector space?

NO!

This is because X can never be the zero vector (since $A \cdot Z \neq B$, where B is non-zero vector), this means that the resultant vector space wouldn't even include the zero vector, so there is no way that X can be a vector space.

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Computing the Nullspace

Basically we're going to be solving for X in $AX=0$.

This involves doing elimination operations, however this time, we cannot stop if any of the pivot elements become zero. And note that via these elimination operations, $C(A)$ changes but $N(A)$ always remains the same.

Example,

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Now pivot element A_{22} is zero, so we look below it to find a non-zero element but we don't. This actually tells us that row 3 is a combination of the previous row or that it is a dependent row. But we don't stop to realize that and instead move on to the next pivot which is A_{23} .

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

And this would be our matrix U which is not exactly an upper triangular matrix (because, you know, its a rectangle) but it is in something called the **echelon form**, this refers to a form where all the non-zero and zero elements of the matrix can be separated by a staircase-like structure.

Rank of the Matrix

In this process of elimination, we required two pivots, i.e. A_{11} and A_{23} . And this is called the rank of the matrix.

So rank of matrix A = 2.

Rank of Matrix = Number of Pivots required

Solving $UX = 0$

Now we can solve the new equation that is $UX = 0$.

For this, we can use the concept of back-substitution.

Prof. Strang uses the term “separating out the pivot columns from the free columns” to describe this.

Pivot columns refers to those columns which contain pivot elements and the rest are free columns. So for matrix A, pivot columns are 1 & 3, free columns are 2 & 4.

Why are these columns called free?

Note that each column is associated with a value in the X vector which is of order 4×1 .

Now **for the free columns, we can substitute any value for the corresponding unknown in X without it affecting anything** and then we can proceed to solve for the remaining unknowns via back-substitution.

So for the given example, we can substitute any random value for x_2 and x_4 .

Say $x_2 = 1$ and $x_4 = 0$

And after back-substitution we get the following. equations,

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 0$$

$$2x_3 + 4x_4 = 0$$

By substituting $x_2 = 1$ and $x_4 = 0$ in these equations we get the value of X as,

$$X = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

You can immediately verify this as its $-2 * \text{col1} + 1 * \text{col2}$.

Ofcourse now, any multiple of X will also satisfy $AX=0$, so they also belong to the nullspace which is a 2D line in 4D space.

Now do the reverse with the value of the free variables, $x_2 = 0$ and $x_4 = 1$. And we get,

$$X = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$2 * \text{col1} - 2 * \text{col3} + 1 * \text{col4} = 0$$

Now we're ready to define the whole nullspace.

These two solutions of X are special solutions because we took special values for the free variables, i.e. kept only one free variable equal to one and the rest equal to zero. There is one special solution per free variable.

Now we can clearly figure out that all linear combinations of these special solutions belong in the nullspace.

That is,

$$N(A) = N(U) = c \cdot \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \cdot \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

We know that the number of pivot columns determines the number of free variables and number of pivot columns are indicated by their rank, so we can determine the following relation,

For a matrix A with m rows and n columns and rank r,

$$\text{Number of free variables} = n - r$$

Reduced Row Echelon Form (rref)

Cleaning up the Echelon Form even more. **Has zeroes above and below the pivot elements and all the pivots are 1s.**

In the example, for matrix U, we got row 3 as all zeros. This actually means that that row is dependent on other rows or is a combination of other rows and the elimination process discovered this fact.

$$U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

U_{13} is above pivot element U_{23} , so we eliminate it,

$$R_1 \rightarrow R_1 - R_2$$

$$U = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot U_{23} can be made 1,

$$R_2 \rightarrow R_2/2$$

$$U = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R, \text{ Reduced Echelon Form}$$

What is the advantage of R?

It immediately tells you that columns 1 & 3 are the pivot columns and which are the pivot elements.

The pivot rows and columns form an identity matrix.

Here because there's two pivot elements in two columns, we get a 2x2 identity matrix

$$\begin{bmatrix} R_{11} & R_{13} \\ R_{21} & R_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The free rows and columns also form a similarly important matrix, call it F.

$$F = \begin{bmatrix} F_{21} & F_{24} \\ F_{31} & F_{34} \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$$

It might not be easy to recognize but the negative of this matrix F is exactly what you get as the values of the pivot variables in the Special Solutions of A.

[Also do note that the solutions of $AX=0$ are the same as that of $RX=0$]

The special solutions were,

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \& \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

The pivot variable values are,

$$x_1 = -2, 2$$

$$x_3 = 0, -2$$

If you multiply this collection of numbers of -1, you will get exactly matrix F.

Why is this so?

Rewrite R assuming that the pivot column values, matrix I comes first followed by whatever's in the free columns, matrix F. And below them, there are the many possible rows of zeros.

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

[Note: This I & F matrix blocks may not be distinct all the time and they may be partly mixed with each other if the pivot and free columns are not sequentially separated.]

The dimensions of this block-type matrix are as follows,

I = pivot rows x pivot columns

$$= r \times r$$

F = free rows x free columns

$$= (n - r) \times (n - r)$$

Nullspace Matrix

N = Columns that are the Special Solutions of $AX=0$

That is, **$RN = 0$**

$$N = \begin{bmatrix} -F \\ I \end{bmatrix}$$

We can solve for $RX = 0$ as,

$$\begin{bmatrix} I & F \end{bmatrix} \begin{bmatrix} x_{\text{pivot}} \\ x_{\text{free}} \end{bmatrix} = 0$$

Multiplying we get,

$$I \cdot x_{\text{pivot}} + F \cdot x_{\text{free}} = 0$$

$$x_{\text{pivot}} = -F \cdot x_{\text{free}}$$

You can see that it satisfies what we have already found. This is why that extra negative sign is present when comparing pivot variable values in the special solutions and in matrix F.

And this is what back-substitution boils down to when you've fully exhausted the reduction process.

Solved Example from Start to Finish

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix}$$

Note that this is the transpose of the previous matrix A..

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix}$$

Next pivot element is zero, look below, A_{32} is non-zero, exchange the two rows.

$$R_3 \iff R_4$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 4 & 4 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 2R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U$$

We got 2 pivot elements, A_{11} and A_{21} . So two pivot columns and one free column and so the rank is two. Also the last two rows are zero which indicates that they were combinations of the previous rows.

It is no coincidence that the number of pivot columns are the same in both A and A^T .

The number of special solutions = Number of free variables = 1.

Solving for nullspace by substituting free variables,

Here, set $x_3 = 1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Therefore, nullspace is given as,

$$X = c \cdot \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Reducing to rref form,

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2/2$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

And you can see that,

$$X = c \cdot \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = c \cdot \begin{bmatrix} -F \\ I \end{bmatrix} = c \cdot N$$

Solving $AX=B$

The only difference here from the previous concept is the RHS is not zero.

Consider the same example that we did before,

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

We already know that the sum of the first two rows is the third row. Say we have 3 values in matrix B as b_1, b_2, b_3 . Then the same logic applies for the b values as well, i.e. the sum of the first two rows must equal the third row, i.e. $b_1 + b_2 = b_3$.

Now we start the elimination process, but because we have non-zero B, we append it to A as the last column to create the augmented matrix, same as how we did in the Gauss-Jordan Method.

In this case, the augmented matrix AB would be,

$$\begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{bmatrix}$$

Now we can perform elimination like we would normally,

$$\begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{bmatrix}$$

If you'll look at the b_3 entry then you'll find exactly what was stated before, that the sum of the first two entries must be equal to the third one.

After back-substitution, the last equation will contain only b values and this represents the condition for solvability, i.e., whether any given B value can be solved by a given system of equations or not.

So if $b_3 - b_2 - b_1 \neq 0$, then that B value cannot be solved.

Say for example,

$$B = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

This is clearly solvable and now we can proceed with solving the remaining two equations to get the value of X , i.e. the 4 unknowns.

Solvability of $AX=B$

We know that $AX=B$ is solvable when B is in $C(A)$.

This same idea can be expressed as,

If a combination of rows of A gives the zero row, then the same combination of entries of B should give zero.

These two statements are equivalent but it may not be that evident right now.

To Find Complete Solution to $AX=B$

First we must find at least one particular solution for X , i.e. $x_{\text{particular}}$ or x_p (for short).

For this Prof. Strang gives a particular method,

Set all free variables to 0.

Solve $AX=B$ for the pivot variables.

Doing this on the continued example we get,

$$x_2 = 0 \text{ and } x_4 = 0$$

And after back-substitution we get,

$$2x_3 = b_2 - 2b_1 = 5 - 2 \cdot 1 = 3$$

$$x_1 + 2x_3 = b_1 = 1$$

Therefore,

$$X_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

Now that we've found a particular solution to X , we must find the solution to the nullspace, i.e. any and all vectors out of the nullspace, $x_{\text{nullspace}}$ (x_n), i.e. set of vectors that solves $AX=0$ and then add them to get the complete solution to $AX=B$.

$$X_{\text{complete}} = X_p + X_n$$

Why does this happen?

We know that,

$$AX_p = B$$

$$AX_n = 0$$

The second equation is true because X_n solves $AX=0$.

Now after adding the two we get,

$$A(X_p + X_n) = B$$

Which is again a valid solution.

This means that if we have one solution, we can add anything from the nullspace solution to it and it will still be a valid solution.

While solving the nullspace, we get special solutions which for this matrix A were,

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

And X_n is given as,

$$X_n = c \cdot \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \cdot \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Therefore, the complete solution is given as,

$$X_{\text{complete}} = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c \cdot \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \cdot \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Now if we try to visualize this solution (which is difficult because it's 4D), we see that we are combining a subspace of $\mathbb{R}^4$ (which is X_n) with a single vector (X_p). Before, the nullspace went through the origin, now it will have been shifted such that it goes through the point described by the vector X_p .

To get X_{complete} , we will combine X_p with all vectors in the subspace of X_n which will give us a subspace-like structure.

Rank & X_{complete}

Say, matrix of order $m \times n$ has rank r .

Then we have the following relations between rank and order.

$$r \leq m \text{ \& } r \leq n$$

Because there can be only one pivot element per row and per column.

Now consider what happens to the solution space, X_{complete} in the following cases,

Full Column Rank: This means that, $r = n$

There are no free variables.

This means that the nullspace consists of only the zero vector as we have no free variables to assign values to.

$$N(A) = \text{Zero Vector}$$

$$X_{\text{complete}} = X_p$$

This means that **the system has a unique solution** and that too in the case that X_p actually exists, i.e. the solvability condition on B is satisfied.

So there is either 0 or exactly 1 solution to this system.

Full Row Rank: This means that $r = m$

So there is a pivot element in each row and there's no rows with full zeros even after elimination.

What does this do to Solvability?

In it, we were required to have atleast one row full of zeros which gave the condition that must be satisfied by values of vector B.

In this case, we have no such rows, so in turn, we have no such condition on the values of B.

This means that we can solve $AX=B$ for every B and a solution will definitely exist, i.e. infinitely many solutions (because of the nullspace).

Also note that we will have $n - r = n - m$ free variables in this case.

Full Rank: $r = m = n$

Obviously this means that it has to be a square matrix.

Since each column has a pivot, there are no free variables, so nullspace is zero vector only and so there is a unique solution.

And since all rows are non-zero, there is no solvability condition on B, so we can solve for every value of B and there will only be one solution for each B.

Summarizing All Cases

The rank tells you everything about the number of solutions.

i) Full Rank

$$r = m = n$$

$$R = I \text{ (R = RREF matrix)}$$

1 Solution

ii) Full Column

$$r = n < m$$

$$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$$

0 or 1 Solution.

iii) Full Row

$$r = m < n$$

$$R = [I \quad F]$$

Infinitely many solutions

iv) Common Case

$$r < m, r < n$$

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

0 (in case the solvability condition fails) or infinitely many solutions (because nullspace is not just zero vector alone).

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Independence, Basis and Dimensions

Suppose A is an $m \times n$ matrix with $m < n$ (i.e. more number of unknowns as compared to number of equations), then there has to be non-zero solutions for $AX=0$. This is because there has to be free columns in the matrix ($n - m$ free variables) which will allow the nullspace to include more than just the zero vector.

Independence

A set of vectors, $X_1, X_2, \dots, X_n$ are linearly dependent if any linear combination with a set of constants gives zero (except for the zero combination, i.e. all constants are zero).

So for them to be linearly independent, the following must hold true for all combinations of c's.

$$c_1x_1 + c_2x_2 + \dots + c_nx_n \neq 0$$

(where, all c's cannot be zero)

A consequence of this is that if any of these vectors are zero, then that means there is no way of independence happening.

In a 2D plane, if we have any two vectors that are not dependent then that can be determined pretty easily.

But what if we have more than 2 vectors to check for independence?

In that case we can directly say that they are dependent.

This is because of the fact stated before, $AX=0$ will have non-zero solutions if there's more columns than rows.

Say there are three non-zero vectors in 2D, then, they can be used to generate matrix A as,

$$\begin{bmatrix} 2 & 1 & 2.5 \\ 1 & 2 & -1 \end{bmatrix}$$

Clearly, number of columns > number of rows, so there will be a non-zero solution for $AX=0$ and so they are dependent.

Now we can form a more formal definition for what classifies independence of a set of vectors.

When vectors $v_1, v_2, \dots, v_n$ are the columns of a matrix A, then they are independent if the nullspace of A is the zero vector.

They are dependent if $Ac = 0$ for some non-zero c.

And how does this relate to the rank of matrix A?

If the vectors are independent, then we should have no free columns, then rank of matrix A must be full column, i.e. n.

And if they are dependent, then the rank has to be smaller than n.

In summary,

Independent $\Rightarrow \text{rank}(A) = n$, no free variables, $N(A) = \{ 0 \}$

Dependent $\Rightarrow \text{rank}(A) < n$, yes free variables, $N(A) = \{ 0 + \text{some more vectors} \}$

Span

If a set of vectors, $v_1, v_2, \dots, v_l$ span a space then that means the space consists of all combinations of those vectors.

This means that the span is the smallest space with those vectors in it.

Using this we can say that the columns of a matrix span its column space.

If the columns of the matrix or the vectors were independent, then they'd have just the "right" amount of vectors in the space that they span. And this leads us to the next concept that is the basis of a space.

Basis

The basis of a space is a sequence of vectors $v_1, v_2, \dots, v_n$, such that,

They are independent.

They span the space.

So telling you the basis of a subspace essentially tells you all that you need to know about it.

Clearly, it is not necessary that there is just one basis for a particular space or subspace.

For example, consider $\mathbb{R}^3$

One basis could be,

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

It's clear why this is a basis, the matrix of them is the identity matrix and the only way to make $AX=0$ for it is if $X=0$.

Another basis can be,

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 8 \end{bmatrix}$$

We should also note that for a basis to span a space of R^n it should consist of n vectors. Don't know exactly why that is but it makes sense (try to visualize it for lower dimensions like try to fill R^2 using just one vector, R^3 using just two vectors and so on).

So to span R^n , its basis of n vectors will form an $n \times n$ square matrix.

And now we can derive another condition for the basis in this special case.

NOTE: You've just fallen into a nice trap!

That second basis is not actually independent. If you construct the matrix using it then row 2 and row 1 are the same and so there is dependency between the rows and because of that there is dependency between the columns.

Fix it by changing vector 2 to something like $[1 \ 5 \ 3]$.

n vectors form the basis for R^n if the $n \times n$ square matrix is invertible.

That is, the determinant is non-zero. This is because for a square matrix, we can compute the determinant and it directly tells us whether there is any dependency in the matrix. If the determinant is zero, then the matrix is dependent, else independent.

Note that this doesn't mean that in an n dimensional space, all bases need to have n vectors. If you have less than n vectors, then that will still be a basis but for a space that is the linear combination of those vectors (provided they all are independent).

Dimension

Given a space, every basis for this space has the same number of vectors.

And this number is called the Dimension of that space.

So if you find a basis for a space and it consists of 6 vectors, then all other bases must consist of exactly 6 vectors and 6 is the dimension of that space.

Solved Example to make everything clear,

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

Do these vectors span $C(A)$?

Yes. By definition.

Are these vectors the basis for $C(A)$?

No. Because they are not independent.

What is the basis for $C(A)$?

Natural answer is columns 1 and 2 because they're the pivot columns for matrix A.

Using this answer we can come to the result of another theorem.

$\text{Rank}(A) = 2 \Rightarrow$ Because there were two pivot elements in the first two columns.

The number of vectors in the basis is also 2.

This also constitutes the Dimension of the space $C(A)$.

Thus, we get the result,

$$\text{Rank}(A) = \dim(C(A))$$

Do note that rank is associated with a matrix and dimension is associated with a space, in this case, the column space of the matrix.

If you know the dimension of the column space of a matrix, then you can form any number of bases for that column space. You just have to ensure that the vectors that you form are made up of linear combinations of the columns of the matrix (this ensures that vectors are present in the column space) and they are independent and the number of such vectors equals the dimension of the column space.

Dimension of the Nullspace

Consider the same example as before. What would be the dimension of the nullspace?

That is, how many vectors does it take to span the nullspace?

It definitely isn't 1.

Try substituting the free variables to get the special solutions for matrix A, which are,

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

These two vectors are independent and they span the nullspace. This means that this is the basis of the nullspace and the dimension of the nullspace is 2.

This leads us to the next result,

The dimension of the nullspace is the number of free variables of the matrix which is also related to the rank of the matrix, r , as follows,

$$\dim(N(A)) = \# \text{free variables} = n - r$$

Because rank essentially counts the number of pivot columns, so what's left are the free columns which contain 1 free variable each and this number is $n - r$.

The Four Fundamental Subspaces

[Assume that matrix A is $m \times n$, $r = \text{Rank}(A)$]

1. Column Space

$$C(A) \text{ in } R^m$$

The basis for the column space are the pivot columns of A .

$$\dim(C(A)) = r$$

2. Null Space

$$N(A) \text{ in } R^n$$

The basis for the null space are the special solutions of A , i.e. those obtained when setting one free variable to 1 and remaining free variables to zero.

$$\dim(N(A)) = n - r$$

Cool property discussed later in the course,

$$N(C \cdot D) = N(D)$$

if C is an invertible matrix

This is because if C is invertible then all its columns are independent and don't really affect the nullspace of D .

3. Row Space

All combinations of rows of the matrix A

Ofcourse Prof. Strang likes to work with columns so we can rephrase this as, all combinations of columns of the matrix A^T .

$$\text{Therefore, Row Space} = C(A^T)$$

$$C(A^T) \text{ in } R^n$$

$$\dim(C(A^T)) = r$$

So the dimension of column space = dimension of row space.

Note: Rank of matrix as well as its transpose is the same!

The same concepts that applied to the column space like basis, independence, etc. apply to the row space as well.

Say R is the Reduced Matrix of A which was obtained by using only row operations, then,
 $C(A) \neq C(R)$

That is to say, row operations do not preserve the column space of the matrix. However they do preserve the row space of the matrix!

The basis for the row space is the first r rows of the matrix R [$R = \text{rref}(A)$], not necessarily A , so do note that you need to get to R to get this answer clearly.

This also confirms that the dimension of the row space is also r .

How is the row space being preserved when we go from A to R ?

When we go from A to R , we perform some linear combination of the rows to go from one intermediate step to the next and finally to R .

Now once we're at R , we can reverse the same sequence of operations, so the last performed operation becomes first, plus becomes minus, etc. and these are also linear combinations of rows. And finally we'll end up at A . This means that we can get rows of A using a linear combination of rows of R , so the row space is preserved.

4. Null Space of A^T

$$N(A^T) \text{ in } R^m$$

Typically called the Left Nullspace.

The basis for the Left Null Space are the special solutions of A^T and their count will be equal to the number of free columns.

Therefore,

$$\dim(N(A^T)) = m - r$$

Let y be a vector in $N(A^T)$, then that means,

$$A^T \cdot y = 0$$

To convert this A^T to A , we transpose the whole equation which gives us,

$$(A^T \cdot y)^T = y^T \cdot A$$

This y represents the solutions to $N(A^T)$ and it is on the left which is why it is called the Left Nullspace.

But this is not the general convention, just use the one before, $A^T \cdot y = 0$

How to compute the Left Nullspace?

For this we'll need the matrix $R = \text{rref}(A)$ and we also need to know a single matrix which can express all the operations that were done to convert A to R .

For this we can use the augmented matrix concept used in the Gauss-Jordan method.

$$[A_{m \times n} \quad I_{m \times m}] \rightarrow \text{Perform row operations} \rightarrow [R_{m \times n} \quad E_{m \times m}]$$

This matrix E represents all the operations performed on A to convert it to R.

$$E \cdot A = R$$

[In the Gauss-Jordan method, A was invertible, so for it, $R = I$ and so, $E \cdot A = I$

This just meant that $E = A^{-1}$]

Example,

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

Solving to get E via augmented matrix $[A \ I]$,

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1; R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow -R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & 1 & 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, E = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Now we can say that,

$$E \cdot A = R$$

$$\begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Note that we have already determined that the dimension of left nullspace is $m - r$, here $r = 2$ and $m = 3$, therefore dimension = $3 - 2 = 1$, i.e. the basis for the left nullspace will consist of just one vector.

The left nullspace is the combination of rows that gives us the zero row.
 And clearly the third row of E is the answer for it, -1 of the first row and 1 of the third row = 0 row
 (see that the third row of R is all zeros!). And this is the basis for the left nullspace!

Note: To visualize what the size of the components are in subspaces, like $C(A)$ in $\mathbb{R}^m$. Just write down an example matrix and try to figure out what the column space, nullspace, etc. will look like (not graphically, just in matrix/vector form).

Matrix Space

Same as a vector space, except the space consists of matrices.

A vector space was defined on the closure over operations of addition of vectors and multiplication by scalars, i.e. linear combination of vectors.

Clearly, this same property can be followed by matrices as well.

Say we define a matrix space, M, as the space of all 3x3 matrices and it is indeed closed over a linear combination of any matrices in the matrix space.

What could be a subspace of M?

- All upper triangular matrices (3x3 ofc), U.
- All symmetric matrices, S.
- The intersection of two subspaces is also a subspace, so symmetric and upper triangular => diagonal matrices.

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What could be a basis for M?

The obvious answer is a collection of 9 matrices where each position in the 3x3 matrix is set to 1 and all other positions are set to zero.

So the dimension of M is 9 because the basis consists of 9 matrices.

What would be the basis for symmetric matrices which we have seen is a subspace of M?

It would consist of 6 matrices from the basis of M, the one where the diagonal elements are set to 1 (3 matrices) + matrices where one of the positions in the upper triangular region is set to 1 (3 matrices, and this will automatically tell us what's in the lower triangular region).

So the dimension of all 3x3 symmetric matrices is 6.

Similarly, the basis for all upper triangular matrices is also the same and its dimension is also 6.

Recall that, $S \cup U \neq \text{subspace}$

Instead we consider, $S + U$

Which is a matrix space where each matrix consists of the sum of a matrix from S and a matrix from U. Recognize that this is indeed a subspace of M because this subspace would literally generate M entirely.

$$\dim(S + U) = 9$$

So now we have a couple of dimensional values using which we can derive an interesting relation,

$$\dim(S) = 6$$

$$\dim(U) = 6$$

$$\dim(S \cap U) = 3$$

$$\dim(S + U) = 9 = 6 + 6 - 3$$

$$\dim(S + U) = \dim(S) + \dim(U) - \dim(S \cap U)$$

A wacky example of a subspace is in differential equations.

Consider the following,

$$\frac{d^2y}{dx^2} + y = 0$$

A set of solutions to this equation are $\cos x$ & $\sin x$

There exists many more solutions to this equation but it is the case that you can get all of them via $\cos x$ and $\sin x$.

$$c_1 \cdot \sin x + c_2 \cdot \cos x$$

Now none of these look like vectors, they're functions and we're far away from subspaces but those same concepts still apply, we can still add them, multiply them by constants and take linear combinations. The solution to this differential equation is essentially like finding the nullspace of matrix A in $AX=0$ and the cosine and sine functions form the basis for this nullspace and the dimension of this subspace is 2. There are other bases as well such as, e^{ix} and e^{-ix} but the dimension is always 2.

This is how dimensions and bases are a topic that transcends Linear Algebra into other domains as well.

Rank 1 Matrices

Rank 1 means that each row is dependent on the "first" row and each column is dependent on the "first" column.

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}$$

$$\dim(C(A)) = \dim(C(A^T)) = \text{Rank}(A) = 1$$

We can rewrite A as,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot [1 \quad 2 \quad 5]$$

First Column * First Row

Statement is that every rank 1 matrix A can be written as a product of some row of A and some column of A.

$$A = C * R$$

These rank 1 matrices form the basic building blocks for a matrix.

For example, for a rank 4 matrix A of order 5x17, we can build A as a combination of rank 1 matrices. And we would require 4 such rank 1 matrices because the rank of A is 4.

Sum of two rank r matrices most often will have rank greater than r.

And so subset of rank n matrices are not subspaces. Same goes for subset of rank 1 matrices.

Say we have a vector V in R^4 that looks like,

$$V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \text{ such that } v_1 + v_2 + v_3 + v_4 = 0$$

Is it a subspace of R^4 ? Yes, because you can multiply this by any scalar and still get a vector whose components will add up to zero and same goes for adding to another vector that follows the same property.

What is the dimension of this subspace?

For that we need to know the basis and before that we can try to systematically connect this to the concept of nullspace.

For what matrix A, is this the nullspace for?

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = 0$$

Now in this case it is clear that the rank of A is 1 and so,

$$\dim(N(A)) = n - r = 4 - 1 = 3$$

And the basis for this subspace can be found by using the method of substituting the free variables, i.e. the special solutions of $N(A)$,

$$s_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, s_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, s_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

This is how you can systematically identify the dimension and basis of a subspace by mapping it to the four fundamental subspaces of a matrix.

For practice, following are the remaining fundamental subspaces for A,

$$C(A^T) = R^1$$

$$C(A) = R^1$$

$N(A^T) = \{0\}$, a single column can be made zero by only multiplying $c = 0$

Note that $\dim(N(A^T)) = 0$, because it only consists of the zero vector

The basis is empty which is why the dimension is zero.

Now consider the dimensions of each subspace and how they relate to the matrix A,

$$\dim(N(A)) + \dim(C(A)) = 3 + 1 = 4 = n$$

$$\dim(N(A^T)) + \dim(C(A^T)) = 0 + 1 = 1 = m$$

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Graphs

$$G = (V, E)$$

where, V = set of vertices

E = set of edges

You know the rest.

Directed and undirected graphs.

Incidence Matrix

Similar to adjacency matrix but rows represent edges of the graph and columns represent nodes of the graph.

The definition for Incidence Matrix elements can be given as follows,

Let I = incidence matrix

for each edge (u, v) with edge number i in G :

$$I(i, j) = 0 \text{ if } j \neq u \text{ \& } j \neq v$$

$$= 1 \text{ if } j = v$$

$$= -1 \text{ if } j = u$$

Try it out with an example to make things clear but basically for each edge, the edge's outgoing vertex will have -1 and the edge's incoming vertex will have +1 and all other vertices will have zero.

If there's a loop in the graph, then that would correspond to linearly dependent rows in the incidence matrix.

Also note that this matrix is a very sparse matrix as each row only contains 2 non-zero values and everything else is zeros.

Given the following incidence matrix, compute its Nullspace,

$$A \cdot x = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

It might not be that clear immediately but we can get the nullspace quickly by direct back-substitution instead of solving the traditional way.

After back-substitution we get,

$$\begin{bmatrix} -x_1 + x_2 \\ -x_2 + x_3 \\ -x_1 + x_3 \\ -x_3 + x_4 \end{bmatrix} = 0$$

What would make this zero?

It's clear that all elements of X set to 1 would make it zero. So, we get one X as,

$$X = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

And what would be the basis?

It would be just this one vector, so $\dim(N(A)) = 1$.

Applied Math & Networks

This can be mapped to the concept of potentials between nodes in an electrical network, where each value of x is a potential at one of the 4 nodes in the network.

And the quantity AX, represents the total current. So if $AX=0$, then that means there is zero current and we now know that $AX=0$ when all components of X are set to the same constant (nullspace). The physical interpretation of this is that there is zero current when all nodes in the network are at the same potential, so there is no potential difference for the current to flow!!!

There is more to the electrical network analogy here, we can see that $\text{rank}(A) = 3$, any 3 columns are independent of each other. So essentially the 4th column can be ignored, in which case, $x_4 = 0$ which can be equated to grounding node 4, setting its potential to zero. And in an electrical network, by grounding a node we can measure the potential at the other nodes.

Taking this further to currents and Kirchoff's Current Law,

In, $Ax = 0$, A consisted of nodes and $x_1, x_2, \dots$ represented potentials and $Ax=0$ computed the potential difference between the nodes.

However, in $A^T y = 0$, $y_1, y_2, \dots$ represents the currents that are on the edges and equation $A^T y = 0$ represents Kirchoff's Current Law (KCL). That's why we need to compute $N(A^T)$ as well.

Also to visualize wtf is actually going on, draw the graph and apply back-substitution on it. Assuming that the current on the edges corresponds to the y components, you can see how the back-substituted equations are the same as those you would get if you had applied KCL on the graph.

After applying KCL on the graph, you will get the following equations representing KCL and the two bases for $N(A^T)$.

Equations,

$$-y_1 - y_3 - y_4 = 0$$

$$y_1 - y_2 = 0$$

$$y_2 + y_3 - y_5 = 0$$

$$y_4 + y_5 = 0$$

Bases,

$$\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

The graph consists of two independent loops (current loops, not typical Computer Science graph cycle), the total current in each loop must be zero which is what gives us these two bases. It also consists of one large loop that spans the entire network but sending current along it would give us a y vector that is the sum of the two bases that we have already found so it's not independent.

If we try to find $C(A^T)$, we know that dimension = 3, so we need to find 3 independent columns. Here, we can choose any 3 columns which do not form a loop. For example in the bases for $N(A)$, we can see that, $y_1 - y_2 + y_3 = 0$, so we can pick y_1, y_2, y_4 to be the independent columns.

Using all of this we can derive one more formula for $\dim(N(A^T))$,

$$\dim(N(A^T)) = m - r$$

$$\#loops = \#edges - (\#nodes - 1)$$

because rank=n-1

Note: Independent loop is one that cannot be composed of smaller loops

If we rearrange this equation we'll get a pretty epic result,

$$\#nodes - \#edges + \#loops = 1$$

This is also called Euler's Formula

And it is true for any graph.

To summarize all of this Applied Math stuff recall this,

$e = Ax$ which measures potential differences.

$y = Ce$ which is Ohm's Law, y represents the current, e is potential difference and C is the constant (which is resistance).

$A^T y = 0$ measures the total current in the network, if we apply some external current source, then this changes to, $A^T y = f$, where f is the external current source.

Substituting values of y and then e in $A^T y = f$ we get,

$$A^T C A x = f$$

And this is the basic form of any applied math in linear algebra, it always boils down to a balance equation.

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Orthogonality

For starters, know that the row space and the nullspace are orthogonal. Also, the column space and left nullspace are orthogonal.

Orthogonal means perpendicular, i.e. each vector in row space is perpendicular to each vector in the nullspace and so on.

Test for Orthogonality

For two vectors x and y , they are orthogonal if their dot product is zero, i.e.,

$$x^T \cdot y = 0$$

Proof:

For this we need to connect the Pythagoras Theorem to this dot product result. Why does the square of sum of sides of a right angled triangle equals the square of the hypotenuse and why is the dot product of the two non-hypotenuse side vectors of that same triangle equal to zero?

According to Pythagoras Theorem,

$$|x|^2 + |y|^2 = |x + y|^2$$

Key thing to note is that for a vector x ,

$$|x|^2 = x^T \cdot x$$

So we can rewrite that equation as,

$$\begin{aligned} x^T \cdot x + y^T \cdot y &= (x + y)^T \cdot (x + y) \\ &= x^T \cdot x + x^T \cdot y + y^T \cdot x + y^T \cdot y \end{aligned}$$

$$x^T \cdot y + y^T \cdot x = 0$$

The dot product is commutative (atleast for real vectors), so,

$$2(x^T \cdot y) = 0$$

$$x^T \cdot y = 0$$

Note: The zero vector is orthogonal to every other vector because its dot product with any vector is zero.

Orthogonality of Subspaces

A subspace S is orthogonal to subspace T if every vector in subspace S is orthogonal to every subspace in T.

In 3D space, are the XY and XZ planes orthogonal?

Very tempting to say yes but it's no!

You can pick a vector in XY that's, say, 45 degrees from the origin and then you can pick a vector in XZ so that its angle at the origin is something else, like 90 perhaps, so that their actual angle is not 90. Requires some visualization but you'll get it. More concrete example is the X axis itself. It belongs to both planes but clearly a vector cannot be orthogonal to itself (unless it's the zero vector) so the planes are not orthogonal.

So for two subspaces to be orthogonal, they cannot have any intersection other than the zero vector at most. So two lines are orthogonal if the only point they meet at is the origin. A line and a plane are never orthogonal (because reasons).

Rowspace is Orthogonal to Nullspace

Why?

Nullspace X is such that $AX=0$

And rowspace is the linear combination of rows of A.

Now if we write $AX=0$ in matrix form, it'll become clear as to why they are orthogonal.

$$\begin{bmatrix} \text{Row 1} \\ \text{Row 2} \\ \dots \\ \text{Row } n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Each entry of the zero vector in the RHS is obtained by the dot product of a row of A with X. This means that each row of A is orthogonal to X (every X).

To prove that the rowspace is orthogonal to the nullspace we also need to show that each combination of rows of A is also orthogonal to X.

This can be done as follows,

$$c_1(row_1)^T \cdot X = 0$$

$$c_2(row_2)^T \cdot X = 0$$

where c_1 & c_2 are some constants

$$c_1(row_1)^T \cdot X + c_2(row_2)^T \cdot X = 0$$

$$(c_1 \cdot row_1 + c_2 \cdot row_2)^T \cdot X = 0$$

Thus the rowspace and nullspace are orthogonal.

Columnspace is Orthogonal to Left Nullspace

This is the same proof as the previous one.

The point to note is that the equation for the Left Nullspace is, $A^T X = 0$

This would compute the dot product of the columns of matrix A with X and make it equal to zero, thereby making them orthogonal and the columnspace becomes orthogonal using the same logic as above.

Summary

The rowspace and nullspace were carving up R^n subspace into an r and n - r dimensional subspace, respectively.

Similarly, the columnspace and left nullspace were carving up the R^m subspace into an r and m - r dimensional subspace, respectively.

For example in 3D space you cannot have a rowspace that's a line and a nullspace that's a line as well making them "unorthogonal" because their dimensions would add up to 2 whereas it should have added up to 3.

Thus the rowspace and nullspace are **orthogonal complements** in R^n . This means that the rowspace contains all vectors of R^n that are perpendicular to the nullspace. Same is for columnspace and left nullspace in R^m .

$A^T A$

What is the "solution" to an equation $AX=b$ if we know that there exists no solution?

This situation might seem retarded but it arises many times in real life because we have too many equations but not enough unknowns to solve. In this case we try to do something to the equation so that we can get a "best" solution.

This involves multiplying by A^T on the left side.

So the new equation is,

$$(A^T A) \cdot \hat{X} = A^T \cdot b$$

Note that X has changed to X hat because this equation might have possible solutions which the previous equation didn't.

$A^T A$ provides additional conveniences that its a square matrix always and it is symmetric as well.

It would be very beneficial if it were invertible as well. When would it be invertible?

When the matrix $A^T A$ has full rank which implies that A has full column rank.

This gives us a few more very important relations,

$$N(A^T A) = N(A) = Z \text{ (Zero Vector)}$$

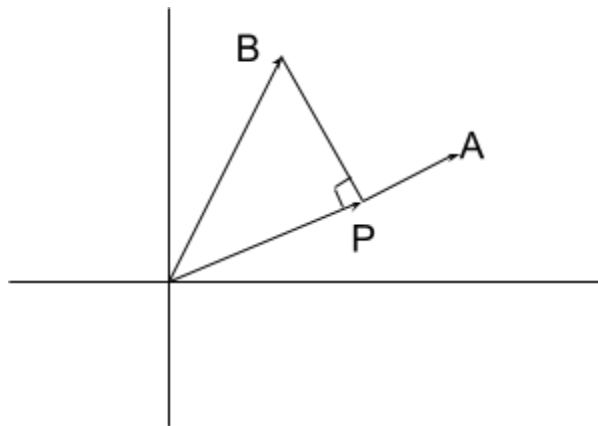
$$\text{Rank}(A^T A) = \text{Rank}(A)$$

To rephrase the invertibility condition in a better way,

$A^T A$ is invertible exactly when $N(A)$ includes the zero vector only, i.e. A has all independent columns, i.e. $\text{rank}(A) = n$.

Projection

Projection of B onto A



Point P is the projection of vector b onto vector a. Orthogonality comes into the picture because the projection of b is perpendicular to a. The projection essentially shows the difference between

the two vectors, i.e. how off were we from vector b to reach vector a, $e = b - p$, e = error

p is some multiple of a because it lies on the same line, $p = x \cdot a$, so to get p we have to find x.

a is perpendicular to e, so by orthogonality equation,

$$a^T \cdot (b - x \cdot a) = 0$$

$$a^T b = x \cdot a^T a$$

$$x = \frac{a^T \cdot b}{a^T \cdot a}$$

$$\begin{aligned} p &= a \cdot x \\ &= a \cdot \frac{a^T \cdot b}{a^T \cdot a} \end{aligned}$$

From this formula and intuition we can determine a couple of things,

If we double vector b , then the projection p also doubles which makes geometrical sense as vector p would be shifted ahead. If we double vector a , then the projection p remains the same as we are projecting b onto a .

Projection Matrix

P = Projection Matrix, p = projection vector, b = vector given as input to projection matrix

$$p = P \cdot b$$

And from the above formula we know that P is,

$$P = \frac{a \cdot a^T}{a^T \cdot a}$$

This is a special matrix, its numerator is $(n \times 1) \cdot (1 \times n) = (n \times n)$ matrix and its denominator is a $(1 \times n) \cdot (n \times 1) = (1 \times 1)$ scalar number.

30/08/21

Properties of Projection Matrix

When you multiply P by vector b , you get projection vector p which is on the line of vector A .

This means that the column space of matrix P is the line containing vector A and its rank is 1.

$$C(P) = \text{Line through } A$$

$$\text{Rank}(P) = 1$$

The rank is 1 because you only need 1 basis to specify a column space that consists of a line and so its dimension is 1 causing rank to be 1.

For more justification recall this statement,

Every rank 1 matrix A can be written as a product of some row of A and some column of A . This is exactly what matrix P is, a is an $n \times 1$ column vector and a^T is a $1 \times n$ row vector, so a column multiplied by a row.

Is P symmetric?

Yes, just take the transpose of P and work out the calculation, it'll come out to be the same as P .

$$P^T = P$$

What happens if you try to take projection again?

Geometrically, you are already at vector p after the first projection, $p = P \cdot b$, so another projection would just leave you at vector p only. No change observed.

Mathematically this means that,

$$p' = P \cdot (P \cdot b) = P^2 \cdot b = p$$

$$\text{Implies, } P^2 = P$$

Note: Remember that the Projection Matrix is symmetric and its square is the same as itself. Very important properties to identify a projection matrix.

Why do we need Projection?

As discussed before, many times we may have a situation where $Ax=b$ has no solution because b is not in the columnspace of A . In that case, we would at least try to get a solution that is as close to b as possible.

For this we'll try to solve, $A \cdot \hat{x} = P$

Where P is the projection matrix of b onto columnspace of A .

Note that here A is a matrix and not a vector as was in the 2D projection example.

Projection for Higher Dimensions

For example, in 3D space we have a plane A and a vector b that is not in the plane, so no solution exists for $Ax=b$, however we can find P by projecting b onto the plane.

Let a_1 and a_2 be the two bases for the plane, now we can represent the plane as the columnspace of matrix A ,

$$A = [a_1 \quad a_2]$$

Now recall that $e = b - p$ essentially represents the error as in how far away b is from its projection onto the plane. And it is very very important that **e is perpendicular to the plane**, i.e. orthogonal.

The projection p belongs to the columnspace of matrix A (because it is in the same plane), so,
 $p = a_1 \cdot \hat{x}_1 + a_2 \cdot \hat{x}_2 = A \cdot \hat{x}$

So we have to find the right combination of columns of A ($\hat{x}$) so that e becomes perpendicular to the plane.

In this example, e should be orthogonal to a_1 and a_2 because they're the bases to the plane.

$$a_1^T \cdot (b - A \cdot \hat{x}) = 0$$

$$a_2^T \cdot (b - A \cdot \hat{x}) = 0$$

This can be represented in matrix form as,

$$\begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix} \cdot (b - A \cdot \hat{x}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^T \cdot (b - A \cdot \hat{x}) = 0$$

Note that you got this same equation in the 2D line case as well, instead of matrix A you had vector a and $\hat{x}$ was just x .

And clearly,

$$e = b - A \cdot \hat{x} \in N(A^T)$$

And this goes with what we already know, $N(A^T)$ is orthogonal to $C(A)$. Since e is orthogonal to $C(A)$, it must mean that e belongs to $N(A^T)$. Also true that the two subspaces are complements of each other wrt to the whole subspace, so this is bound to happen.

Rewriting the matrix form equation we get,

$$A^T \cdot A \cdot \hat{x} = A^T \cdot b$$

$$\hat{x} = (A^T A)^{-1} A^T \cdot b$$

$$p = A \cdot \hat{x} = A \cdot (A^T A)^{-1} A^T b$$

This is also the same as in the 2D case except we're not allowed to divide by $A^T A$ so we write the inverse of it.

And since $p = P \cdot b$, we get,

$$P = A \cdot (A^T A)^{-1} \cdot A^T$$

Don't you dare try to open that inverse parentheses and prove that $P = \text{Identity Matrix!!!}$

You cannot open the parentheses because A is not always a square matrix and because of which A^{-1} doesn't exist. $A^T A$ is square and which is why we can take its inverse but in most cases A will not be (i.e. too many rows and too few columns, many equations and less unknowns).

If A was indeed a square invertible matrix, then in fact P would come out to be equal to I .

What does that mean?

It means that A is an $n \times n$ matrix consisting of n independent vectors in n -dimensional space which means that $C(A)$ is the entirety of $\mathbb{R}^n$. This means that any vector b in n -dimensional space is already covered by $C(A)$ and so there is no error in the projection, i.e. $p = P \cdot b = I \cdot b = b$

The two key properties for P still hold,

P is still symmetric, $P^T = P$

Projection of the projection just leaves us with the same vector, $P^2 = P$

[Try it out yourself and the result is quite easy]

Key point to note:

Even if $A \cdot x = b$ doesn't have a solution, $A^T A x = A^T b$, does.

Least Squares Problem

You are given a collection of data points in 2D where they don't necessarily lie on the same line, but we're supposed to find the best possible line through all these points.

Consider the following example, b is the Y-axis and t is the X-axis and the data points are, (1, 1) (2, 2) (3, 2). Fit a line through these points such that the error is minimized.

Let this minimal error line be given by,

$$b = C + Dt$$

We can connect this to a matrix by substituting the data point values in these equations. We have 3 points, so we get 3 equations.,

$$C + D = 1 \rightarrow (1, 1)$$

$$C + 2D = 2 \rightarrow (2, 2)$$

$$C + 3D = 2 \rightarrow (3, 2)$$

And we can write this in matrix form as follows,

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$\begin{bmatrix} C \\ D \end{bmatrix}$ represents a solution to this system of equations, i.e. a line that fits through all three points but we know that that's impossible. And this is exactly why projection helps, we have a system of equations and there's no solution for it but we would like to get a solution that is as close to the actual solution as possible.

So instead of solving $Ax=b$, we will solve,

$$A^T A \hat{x} = A^T b$$

Revising some concepts

$$P = A \cdot (A^T A)^{-1} \cdot A^T$$

Also, $p = P \cdot b$.

Consider the extreme cases,

b is perpendicular to $C(A) \Rightarrow$ It must belong to $N(A^T) \Rightarrow A^T \cdot b = 0$

$$\begin{aligned} P \cdot b &= A(A^T A)^{-1}(A^T \cdot b) \\ &= A(A^T A)^{-1}(0) = 0 \end{aligned}$$

b is in $C(A) \Rightarrow b = Ax$ (Ax = linear combination of columns of A)

$$\begin{aligned} P \cdot b &= A \cdot (A^T A)^{-1} \cdot A^T \cdot Ax \\ &= A \cdot I \cdot x \\ &= Ax \\ &= b \end{aligned}$$

When we have a matrix A , we're in fact doing two projections of vector b onto A .

First, we're **projecting b onto $C(A)$** which is called the **vector p** and it is given as, $p = P \cdot b$.

Then, we're **projecting b onto $N(A^T)$** which is called the **vector e** and it is given as,

$$e = b - p = b - P \cdot b = b \cdot (I - P)$$

P is a projection matrix and so $I - P$ is also a projection matrix because it too is symmetric and idempotent as well.

Solving the Least Squares Problem

Continuing the same example from before,

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

This is in the form $Ax = b$

And we want to compute the best fitting line through the three points, however there's no such perfect line for it and so the best fitting line will have error and we have to minimize this error which can be given as,

$$\text{Minimize } Ax - b = e$$

But we're concerned with the length of the error so we need the magnitude of e plus we square the error so that the effect of the error is more pronounced, even small errors are given high importance.

$$\text{Minimize } |Ax - b|^2 = |e|^2$$

And for this example where we have three data points, we will get three error vectors which will factor into the final error e .

The consequence of this squaring is that the effect of outliers is greatly pronounced, so even a single outlier can wildly screw up the entire line that we were going to get by solving the system. So in such cases where outliers matter, the least squares method is not directly used.

FYI this is Linear Regression we're doing here.

Now finally we can solve this problem,

$$\text{Find } \hat{x} = \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} \text{ and } p$$

We solve for x via the equation,

$$A^T A \cdot \hat{x} = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$3C + 6D = 5$$

$$6C + 14D = 11$$

These are called the normal equations.

Note that we could have solved these equations using the least squared error logic discussed which involves minimizing e and that deals with calculus, taking the partial derivative wrt C and

D to get two equations. These two equations will be the same as the normal equations that we obtained here.

Now we can solve the normal equations using regular elimination and we get,

$$D = 1/2, C = 2/3$$

Thus the best line through the three points is,

$$2/3 + 1/2 \cdot t$$

Substituting the values of t in this line at the data points gives us the error terms,

$$\begin{aligned} e_1 &= b_1 - \text{Point 1 (t=1)} \\ &= 1 - (2/3 + 1/2 \cdot 1) \\ &= -1/6 \end{aligned}$$

Similarly,

$$\begin{aligned} e_2 &= 2 - (2/3 + 1/2 \cdot 2) \\ &= 1/3 \end{aligned}$$

$$\begin{aligned} e_3 &= 2 - (2/3 + 1/2 \cdot 3) \\ &= -1/6 \end{aligned}$$

Therefore,

$$e = \begin{bmatrix} -1/6 \\ 2/6 \\ -1/6 \end{bmatrix}, p = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}$$

[p is obtained while calculating e itself, Point 1(t=1), Point 2(t=2), Point 3(t=3) are the p terms]

And clearly $b = e + p$

You can verify your solution by noting that e should be perpendicular to p as well as to anything in $C(A)$. So, compute those dot products and they should come out to zero.

And finally via the other formula,

$$p = A \cdot \hat{x}$$

Invertibility of $A^T A$

It was stated that $A^T A$ is invertible only when all columns of A are independent.

Prove this.

Suppose,

$$A^T A \cdot x = 0$$

In this case we'd like to show that x has to be the zero vector which means that $N(A^T A)$ is just the zero vector which only happens when there are no free variables, i.e. full column rank, i.e. all columns are independent.

Idea: Pre-multiply by x^T on both sides

$$x^T \cdot A^T A \cdot x = 0$$

$$(A \cdot x)^T \cdot (A \cdot x) = 0$$

The dot product of a vector and its transpose is zero only when the vector is zero!

This means that $A \cdot x = 0$ and because all columns of A are independent, $N(A)$ is just the zero vector and so $x = 0$.

Important Note:

Columns of a matrix are definitely independent if they are perpendicular unit vectors (this unit vector condition excludes the zero vector).

An example is,

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

And the matrix of these vectors is the identity matrix so $A^T A = I$ itself.

These perpendicular unit vectors are also called orthonormal vectors.

Do remember that they are always independent.

Another example for orthonormal vectors are,

$$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

Both are unit vectors and independent and it's difficult to visualize but they're orthogonal as well.

01/09/21

Orthonormal Vectors & Orthogonal Matrices

Two **unit** vectors q_i and q_j are orthonormal if,

$$\begin{aligned} q_i^T q_j &= 0 \text{ if } i \neq j \\ &= 1 \text{ if } i = j \end{aligned}$$

0 part ensures orthogonality (ortho) and 1 part ensures that they are unit vectors (normal).

Consider a matrix Q that consists of orthonormal vectors q_1 to q_n in each column.

$Q^T Q$ is a very special result as follows,

$$\begin{aligned}
Q^T Q &= \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} \cdot \begin{bmatrix} q_1 & \dots & q_n \end{bmatrix} \\
&= \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \dots & q_1^T q_n \\ \vdots & \vdots & \vdots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \dots & q_n^T q_n \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
Q^T Q &= I
\end{aligned}$$

Note: A matrix consisting of orthonormal columns is called an orthogonal matrix only when it's a square matrix.

If Q is square matrix, then $Q^T Q = I$ tells us that,
 $Q^{-1} = Q^T$

Some examples for matrix Q,

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Check that each column vector's length is 1 (unit) and the dot product with any other column is 0 apart from itself.

Also note that matrix 3 is just matrix 2 with theta = 45 degrees.

$$Q = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Part of a famous orthogonal matrix series, Ademar's Orthogonal Matrices.

Orthonormal Basis

The columns of an orthogonal matrix gives you an orthonormal basis for the column space of the matrix. Why?

Because orthonormal vectors are independent of each other.

To visualize this better consider the following,

For them to not be independent, you have to be able to make one vector using linear combinations of the other vectors. But this vector is orthogonal to all the remaining vectors so there is no way you can make it by combining the other vectors in any possible way. This basically proves that each orthogonal vector in an orthogonal matrix is independent of each other.

Projection of Q

Project orthogonal matrix Q onto its column space, then the projection matrix is given as follows,

$$P = Q(Q^T Q)^{-1} Q^T$$

You can see the main advantage of having orthogonal matrices right here. Transpose and inverse are the same!

$$P = Q \cdot I \cdot Q^T = Q \cdot Q^T$$

We do not reduce the RHS to I because it is not necessary that Q is a square matrix.

If Q is square, then definitely $P = I$, because all columns of Q are independent and $C(Q)$ would be the entire space. So, any vector b would always be present in it hence there's no error in projection.

You can check that the properties of projection matrices hold true for $Q \cdot Q^T$. It is symmetric and its square is also $Q \cdot Q^T$ (idempotent).

$$\begin{aligned} (Q \cdot Q^T) \cdot (Q \cdot Q^T) &= Q \cdot (Q^T \cdot Q) \cdot Q^T \\ &= Q \cdot I \cdot Q^T \\ &= Q \cdot Q^T \end{aligned}$$

You can see that all the complicated equations are becoming simpler when we have Q as the matrix.

Before we had, $A^T \cdot A\hat{x} = A^T \cdot b$

Now we have,

$$Q^T \cdot Q\hat{x} = Q^T \cdot b$$

$$I \cdot \hat{x} = Q^T \cdot b$$

$$\hat{x} = Q^T \cdot b$$

This basically says that,

$\hat{x}_i = q_i^T \cdot b$, which is a really important equation in mathematics.

All of this should motivate the importance of having orthogonal matrices Q. And if we do have regular matrices, we should try to make them orthogonal so that our calculations become simpler.

Gram-Schmidt

Start with independent vectors as columns of a matrix and make the matrix orthogonal.

1. Start with two vectors a and b
2. Make them orthogonal, i.e. get two orthogonal vectors A and B.
3. Make them orthonormal, i.e., divide A and B by their lengths.

Making a vector Orthogonal

You have vector a and it doesn't violate any conditions of orthogonality, so you keep it as it is, i.e. $A = a$

Now you add vector b to the mix and orthogonality gets violated, so you have to modify vector b to make it B.

It takes a bit of visualization but the vector B that's orthogonal to A, is the error term e that's associated with the projection of b onto A.

The error term that we had defined as,

$$e = b - p = B$$

And after substituting the value of p we get,

$$\begin{aligned} B &= b - \frac{A \cdot A^T}{A^T A} \cdot b \text{ (From what we had learned earlier)} \\ &= b - \frac{A^T \cdot b}{A^T A} \cdot A \text{ (This rearrangement is actually valid and it makes calculations easier)} \end{aligned}$$

Why does this make calculation easier => Check dimensions of the two equations.

$A \cdot A^T \Rightarrow (n \times 1) * (1 \times n) \Rightarrow (n \times n)$ matrix

$(A \cdot A^T) \cdot b \Rightarrow (n \times n) * (n \times 1) \Rightarrow (n \times 1)$ vector

1 Vector-Vector + 1 Matrix-Vector Multiplication

$(A^T \cdot b) \Rightarrow (1 \times n) * (n \times 1) \Rightarrow (1 \times 1)$ scalar

$(A^T \cdot b) \cdot A \Rightarrow (1 \times 1) * (n \times 1) \Rightarrow (n \times 1)$ vector

1 Vector-Vector + 1 Scalar-Vector Multiplication

Now all we have to do is divide the orthogonal vectors by their lengths to make them orthonormal.

But what if there are 3 vectors?

A and B we know, but how to calculate C, the third orthogonal vector.

Repeat the same that we did for B, i.e. we removed the component of b in A's direction (its projection) to get B, now we remove the component of c in the directions of A and B both.

Therefore,

$$C = c - \frac{A^T c}{A^T A} \cdot A - \frac{B^T c}{B^T B} \cdot B$$

And C will be orthogonal to both A and B.

This also means that if $C = 0$, then C can be expressed as a linear combination of both A and B . Because if we remove the component of c in the directions of A and B and we're left with nothing $\Rightarrow C = 0 \Rightarrow C$ is dependent on A and B .

Example problem for two vectors,

$$a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$A = a$$

$$\begin{aligned} B &= b - \frac{A^T b}{A^T A} \cdot A \\ &= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ B &= \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

You can confirm that A and B are orthogonal by verifying that $A \cdot B = 0$

Therefore matrix Q is given as,

$$q_1 = A/\sqrt{3}, q_2 = B/\sqrt{2}$$

$$Q = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$

Take time to understand that $C(Q) = C(M)$

[M = the starting matrix with columns as vectors a & b]

We're staying in the same space, just choosing vectors that are perpendicular to each other.

Just like in elimination, we decomposed all the row operations into a matrix L such that, $A = LU$.

For Gram-Schmidt, we will decompose all the orthogonalization and normalization operations into a matrix and denote this matrix as R and $A = QR$.

We can write this out as,

$$A = [a \quad b] = \begin{bmatrix} q_1 & q_2 \end{bmatrix} \cdot \begin{bmatrix} a^T \cdot q_1 & b^T \cdot q_1 \\ a^T \cdot q_2 & b^T \cdot q_2 \end{bmatrix}$$

Turns out that,

$$a^T \cdot q_2 = 0$$

This is because all orthonormal vectors obtained (q 's) are such that they are orthogonal to all previous vectors.

This will remain true in the general case as well, which means that in matrix R the entire lower diagonal portion becomes zero, **making R a lower triangular matrix**. This makes this quite similar to the $A = LU$ case, because L was also a lower triangular matrix. So that's quite satisfying to see!

To find R, we can use the following logic,

$$A = QR$$

$$Q^T A = Q^T QR \text{ [Pre-multiplying by } Q^T]$$

$$R = Q^T A [Q^T Q = I]$$

Combining all of this together we can write the Gram-Schmidt Algorithm,

Given n vectors, $a_1 \dots a_k$

for $i = 1 \dots k$,

1. Orthogonalization, $\tilde{q}_i = a_i - q_1^T \cdot a_i \cdot q_1 - \dots - q_{i-1}^T \cdot a_i \cdot q_{i-1}$

2. Test for linear dependence, if $\tilde{q}_i = 0$, then stop.

3. Normalization, $q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$

Process terminates early if any of the a_i vectors are dependent.

Note that for orthogonalization of i^{th} vector, we only subtract components from q_1 upto q_{i-1} vectors.

Also, while projecting we divide by $q_i^T \cdot q_i$. However because these vectors are already orthonormal, this will come out to be 1 and that's why we don't include it in the equation for orthogonalization above.

15/10/21

[Bonus stuff from Vishwanath sir's lecture ppts]

Solving $AX = B$ using QR Factorization

The process of decomposing matrix A as $A = QR$ is called QR factorization.

Now using this we can develop a computationally efficient method for solving $AX = B$ as follows,

Perform QR factorization, $A = QR$

$$Ax = b \implies Q^T QRx = Q^T b \implies Rx = Q^T b$$

Compute $Q^T b$

Solve $Rx = Q^T b$

[This is straightforward because R is already upper triangular so elimination is already done]

So we can directly use back-substitution to solve the final equation.

Overall you can see how this provides us an algorithmic approach to solving this problem instead of doing the elimination process ourselves using row operations that are expensive to do.

So just use the Gram-Schmidt algorithm to find Q, then find R to get $A = QR$, then do the steps in the method described above.

Moore-Penrose Pseudo Inverse

We know that $A^T A$ has inverse when A has full column rank, i.e. all columns of A are independent of each other.

Then the Moore-Penrose Pseudo Inverse is given as,

$$A^\dagger = (A^T A)^{-1} A^T$$

When A is non-square, then $A^\dagger$ is the left inverse of the matrix and is the regular inverse when A is square.

A matrix can have many left inverses however the pseudo inverse is unique.

We can also find this pseudo inverse via QR factorization.

Because if matrix A has left inverse then its columns are linearly independent (full column rank) and if columns are linearly independent then it means A can also undergo QR factorization.

$$A = QR$$

$$\begin{aligned} (A^T A)^{-1} A^T &= ((QR)^T (QR))^{-1} (QR)^T \\ &= (R^T Q^T Q R)^{-1} (R^T Q^T) \\ &= (R^T R)^{-1} (R^T Q^T) [\because Q^T Q = I] \\ &= R^{-1} (R^T)^{-1} R^T Q^T \\ A^\dagger &= R^{-1} Q^T \end{aligned}$$

Eigen Vectors

Consider a transformation induced by the square matrix A as follows,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, y = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \implies A \cdot y = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

Similarly we can do other transformations on other vectors as,

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies A \cdot x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot x_1$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \implies A \cdot x_2 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \cdot x_2$$

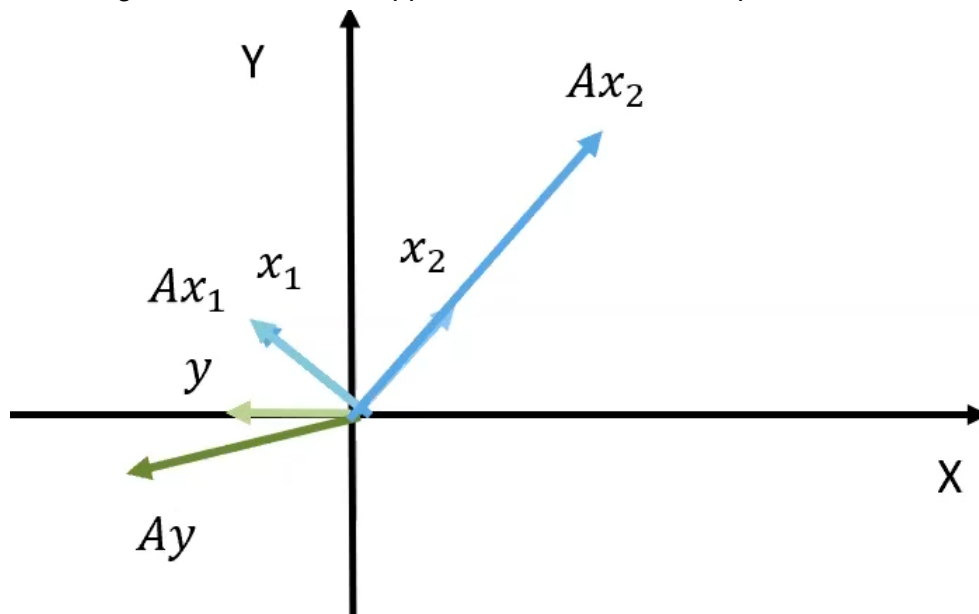
So you can see what's the interesting stuff happening here...

Multiplying A by these vectors gives a scaled form of that vector as the output, i.e. $A \cdot x = c \cdot x$

The geometric interpretation of this is that, for any square matrix A there are certain directions x which when operated on by this matrix A doesn't change the direction of the resultant vector.

Do note that this property will also hold for any vector in the direction of x.

This image illustrates what happened in the above example,



For a square matrix A, certain vectors only get linearly scaled and do not change their direction when operated on by A.

These directions are called **eigenvectors**.

The scaling value the eigenvectors undergo are the **eigenvalues**.

So in the above example, x_1 and x_2 were eigenvectors and 1 and 3 were their respective eigenvalues.

Formal definition

A vector $v \in R^n$ is an eigen vector of an $n \times n$ square matrix A if

$$A \cdot v = \lambda \cdot v$$

Also remember, non-square matrices do not have eigenvalues/vectors.

Can we solve the eigen equation system similar to how we solved $Ax = b$, since λv is just another vector which can be considered as b and x is just v ?

Answer is NO!

Because in the $Ax = b$ case, we knew both A and b and were trying to find x .

However in this case we only know about A and are trying to find both v and λ , so two unknowns and just one equation, not possible.

We can rewrite the equation as,

$$Av = \lambda v$$

$$(A - \lambda I) \cdot v = 0$$

This basically means that,

$$N(A - \lambda I) = v$$

And it also means that the columns of the matrix are not fully independent which is why we're able to get a nullspace in the first place.

This means that the matrix is singular, i.e. its determinant is zero.

$$|A - \lambda I| = 0$$

Goal: Find all sets of (λ_i, v_i) where $(A - \lambda_i I)v_i = 0$, where λ_i is **associated with** the eigen vector v_i

How many such eigenvalues exist for a given matrix A ?

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \implies |A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2 = 0$$

This equation in λ is called the **Characteristic Equation** and the degree of this equation is equal to the order of the matrix A (2x2 here so quadratic equation in C.E).

The roots of the Characteristic Equation will give you the eigenvalues which in this example are 1 and 3.

How to find the Eigenvectors once you have found the Eigenvalues?

Consider the previous example,

$$(A - \lambda I) \cdot v = 0$$

$$\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) \cdot v_{\lambda=3} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Use back substitution (and elimination too) to solve this system which gives you the equation, $v_1 = v_2$.

And this is the eigenvector for eigenvalue = 3

Note that eigenvector is a direction, so we can choose any value for v_1 and it would still be an eigenvector provided $v_2 = v_1$.

So let,

$$v_{\lambda=3} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Similarly we can find the eigenvector for eigenvalue = 1.

Because the C.E is of degree n, it will have n roots and so n eigenvalues.

However!

These n roots can be real/complex and **repeated**//distinct.

$(\lambda - 1)^2 = 0 \implies$ Repeated solution & $\lambda^2 + 1 = 0 \implies$ Complex solution

Note: The determinant of both **A and its transpose** are equal and so the characteristic equation will be the same and therefore the **eigenvalues will also be the same**.

Note: All vectors in $N(A)$ are eigenvectors of A.

This is because,

$Ax = 0 \implies Ax = 0x \implies x$ is eigenvector with $\lambda = 0$

Eigenvectors & Projection Matrix

Projection matrix P is defined wrt a subspace S.

$p = P \cdot b$ will modify all b not in S.

$\forall b \in S, P \cdot b = I \cdot b = 1 \cdot b \implies$ All $b \in S$ are eigenvectors with $\lambda = 1$

$\forall b \perp S, P \cdot b = 0 \cdot b \implies$ All $b \in S$ are eigenvectors with $\lambda = 0$

For the following Projection Matrix for X-Y plane we get the eigenvalues as,

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\lambda \cdot (1 - \lambda)^2 = 0$$

$$\lambda_1 = 0, \lambda_2 = 1$$

There is no third root as such because 1 repeats twice as the eigenvalue.

This is why λ_2 has an **algebraic multiplicity** of 2.

We know that $\lambda = 1$ when b is in S and that b is an eigenvector. How many such independent eigenvectors can we find?

Only 2!

Because the rank of P is 2 and so the last column is dependent on the first two.

Therefore, λ_2 has a **geometric multiplicity** of 2 as we can find 2 independent eigenvectors corresponding to λ_2

Also for λ_1 , the algebraic multiplicity as well as geometric multiplicity are equal to 1.

Geometric Multiplicity $\leq$ Algebraic Multiplicity

Sidenote:

For the rotation matrix,

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The eigenvalues as well as eigenvectors are in the complex plane.

This makes sense because the rotation matrix is supposed to change the vector it is given as input by given angle theta.

But this goes against the definition of what eigenvectors are supposed to do and this is why there's no real eigenvector for R and instead you have to find complex eigenvectors.

Linear Independence of Eigenvectors

If all eigenvalues $\lambda_1, \dots, \lambda_n$ are distinct then all eigenvectors $x_1, \dots, x_n$ are linearly independent.

The proof for this is kinda messy so I just skipped it entirely.

Just take it as fact and roll with it ^_^

If they are not distinct then too independent eigenvectors are also possible (for example the projection matrix example used in the previous section) but it is not a guarantee.

Eigen Decomposition (Spectral Decomposition)

For $n \times n$ square matrix assume that there are n distinct eigenvalues. So there are n independent eigenvectors as well.

We can write all of this as,

$$Av = \lambda v$$

$$Av_1 = \lambda_1 v_1, \dots, Av_n = \lambda_n v_n$$

We can stack the eigenvectors as columns of a matrix and compress the notation as follows,

$$A \cdot \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} A \cdot v_1 & \dots & A \cdot v_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \cdot v_1 & \dots & \lambda_n \cdot v_n \end{bmatrix}$$

And we can further decompose this using more matrix multiplication haxx (column interpretation) as follows,

$$A \cdot \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix}$$

Take some time to justify why this works :(

To summarize,

$$\text{Let } S = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \implies AS = A \cdot \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \cdot \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix} = S\Lambda$$

$$\text{where, } \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix}$$

$AS = S\Lambda$ (all these matrices are square)

Because we assumed n distinct eigenvalues, we know that all eigenvectors are independent and so S^{-1} exists and we can do the following,

$$A \cdot S \cdot S^{-1} = S \cdot \Lambda \cdot S^{-1}$$

$$A = S \cdot \Lambda \cdot S^{-1}$$

This is called **Matrix Factorization** and it is a very useful equation for many applications.

If we instead pre-multiply by S^{-1} , then we get the following,

$$S^{-1} \cdot A \cdot S = S^{-1} \cdot S \cdot \Lambda = \Lambda$$

This is called **Matrix Diagonalization** because it converts matrix A into a diagonal matrix.

This is basically Eigen Decomposition/Spectral Decomposition.

Application: Efficiently Computing Matrix Powers

Say you have to find A^{100} and this is a very computationally heavy operation. So what you do is find the eigenvector x and eigenvalue associated with A .

[Note that A has to be a square matrix]

$$Ax = \lambda x$$

Then you can premultiply by A ,

$$\begin{aligned} A^2x &= \lambda \cdot A \cdot x \\ &= \lambda^2 x \end{aligned}$$

Eigenvectors stay the same and don't mix up, only the eigenvalues grow in power as A grows in power.

Now for diagonalizable matrices (which A is because it can undergo Eigen Decomposition)

$$\begin{aligned} A^2 &= A \cdot A \\ &= S \cdot \Lambda \cdot S^{-1} \cdot S \cdot \Lambda \cdot S^{-1} \\ &= S \cdot \Lambda \cdot \Lambda \cdot S^{-1} \\ A^2 &= S \cdot \Lambda^2 \cdot S^{-1} \end{aligned}$$

It follows that,

$$A^k = S \cdot \Lambda^k \cdot S^{-1}$$

And k^{th} power of the Lambda matrix is just raising each lambda component in it by k.

Symmetric Matrices & Eigen Vectors

$n \times n$ real symmetric matrix A will have,

1. n real eigenvalues (not necessarily distinct).
2. **All eigenvectors are orthogonal to each other.**

Proof for Eigenvalues are real:

A value is real if it has the same value as its complex conjugate

$$z = a + bi, \bar{z} = a - bi$$

If $z = \bar{z} \implies z = a$ & no imaginary part

Few more useful properties,

$$z \cdot \bar{z} = (a + bi)(a - bi) = a^2 + b^2 \implies \text{Real number}$$

If w and z are two complex numbers then,

$$\overline{wz} = \bar{w} \cdot \bar{z} \text{ (check using an example)}$$

Now we can apply these to a vector v and compute its dot product with its complex conjugate,

$$\bar{v}^T \cdot v = \begin{bmatrix} a_1 - ib_1 \\ \dots \\ a_n - ib_n \end{bmatrix}^T \cdot \begin{bmatrix} a_1 + ib_1 \\ \dots \\ a_n + ib_n \end{bmatrix} = (a_1^2 + b_1^2) + \dots + (a_n^2 + b_n^2)$$

This basically tells you that this dot product will always have a real number as output irrespective of whether v is real or imaginary.

Also the dot product is zero only when v itself is zero (because output real number is sum of squares of real numbers).

Onto the actual proof,

We can only assume that A is symmetric and real, everything else can be anything.

$$Av = \lambda v \rightarrow (1)$$

$$\overline{Av} = \overline{\lambda v} \implies \bar{A} \cdot \bar{v} = \bar{\lambda} \cdot \bar{v}$$

A is real so its conjugate is the same

$$\therefore A \cdot \bar{v} = \bar{\lambda} \cdot \bar{v} \rightarrow (2)$$

Premultiply equation 1 with transpose of v conjugate,

$$\bar{v}^T Av = \bar{v}^T \lambda v = \lambda (\bar{v}^T \cdot v) \rightarrow (3)$$

Rearranging equation (3) in a different way we get,

$$\bar{v}^T Av = (A^T \bar{v})^T v = (A \bar{v})^T v$$

From (2) & (3),

$$v^T Av = (\bar{\lambda} \cdot \bar{v})^T v = \bar{\lambda} \cdot (\bar{v}^T v) \rightarrow (4)$$

(3) & (4) are equal now!

$$\lambda(\bar{v}^T \cdot v) = \bar{\lambda} \cdot (\bar{v}^T \cdot v)$$

We have already proven that $\bar{v}^T v \neq 0 \implies \lambda = \bar{\lambda}$

This means that any eigenvalue is real because only then will the conjugate operation not change anything.

Thus all eigenvalues are real for a real symmetric matrix A.

Proof for Eigenvectors are Orthogonal:

x is an eigenvector of A corresponding to the eigenvalue λ_1

y is an eigenvector of A corresponding to the eigenvalue λ_2

$\lambda_1 \neq \lambda_2$ because eigenvalues are assumed to be distinct

$$Ax = \lambda_1 x, Ay = \lambda_2 y$$

After taking into account the fact that A is symmetric,

$$y^T Ax = \lambda_1 y^T x \rightarrow (1)$$

$$x^T Ay = \lambda_2 x^T y \rightarrow (2)$$

Subtracting (1)-(2),

$$y^T Ax - x^T Ay = \lambda_1 y^T x - \lambda_2 x^T y \rightarrow (3)$$

Note that $y^T Ax = \text{Scalar value} \implies$ we can take its transpose without any issue

Also note that $(ABC)^T = C^T B^T A^T$

$$\therefore (y^T Ax)^T = x^T A^T y = x^T Ay \rightarrow [\because A \text{ is symmetric}]$$

$$\therefore (y^T Ax)^T = y^T Ax = x^T Ay$$

$\therefore$ LHS of (3) is becoming zero

$$\lambda_1 y^T x - \lambda_2 x^T y = 0$$

$$y^T x = x^T y \rightarrow [\because \text{dot product is commutative}]$$

$$(\lambda_1 - \lambda_2)y^T x = 0$$

$$\lambda_1 \neq \lambda_2 \implies y^T x = 0 \implies x \text{ and } y \text{ are orthogonal}$$

Thus the two eigenvectors are orthogonal and similarly we can prove that all eigenvectors for such a real symmetric matrix A with distinct eigenvalues are orthogonal.

If the eigenvalues are not distinct then its matrix has a nullspace that isn't just the zero vector.

When the eigenvalues are repeated, then the eigenvectors associated with each repeated eigenvalue span a subspace and this subspace would be orthogonal to the other eigenvectors/eigenspaces (of repeated eigenvalues).

For example, in the projection matrix case, the eigenvalue 1 was repeated twice and we were able to find two independent eigenvectors that spanned the entire column space of the projection matrix. So, this subspace of two eigenvectors associated with eigenvalue 1 will be orthogonal to the eigenvector associated with eigenvalue 0. And we know this is true because eigenvalue = 1 $\Rightarrow$ vector b is in the column space of P (or A) and eigenvalue = 0 $\Rightarrow$ vector b is orthogonal to the column space of P (or A).

[Note: If eigenvalue 0 was also repeated twice, then we would have to find its subspace and then both subspaces associated with eigenvalue 1 and 0 will be orthogonal to each other].

Adding Eigen Decomposition to the mix

Putting all of this together with the concept of Eigen Decomposition,

The matrix S is the stacking of Eigenvectors of A and if A is real, symmetric with distinct eigenvalues then all eigenvectors would be orthogonal to each other.

Therefore matrix S becomes matrix $Q \Rightarrow$ Orthogonal Matrix!

$$A = S\Lambda S^{-1}$$

$$A = Q\Lambda Q^{-1} = Q\Lambda Q^T$$

If eigenvalues are repeated, then we can choose eigenvectors for all the distinct eigenvalues and they'll be orthogonal. For the eigenvectors associated with the repeated eigenvalues we can just pick vectors from the nullspace of the matrix A .

Because as we know, all vectors in the nullspace of the matrix A are eigenvectors.

Every Symmetric Matrix can be factorized as $A = Q\Lambda Q^T$ with real eigenvalues Λ and orthonormal eigenvectors in the columns of Q .

It's a bit difficult to visualize how we got here but...

$$A = Q\Lambda Q^T = \sum \lambda_i \cdot q_i \cdot q_i^T$$

Just expand each matrix and apply tricks related to matrix multiplication like column and row interpretation and you should finally reach here.

Note that $q_i \cdot q_i^T$ is a rank 1 matrix!

Therefore a symmetric matrix A can be expressed as a sum of Rank 1 Matrices

Also note that this rank 1 matrix looks quite similar to the projection matrix,

$$P_i = \frac{q_i q_i^T}{q_i^T q_i} = q_i q_i^T$$

There's a reason why this is happening.

When we operate vector b by a matrix A , i.e. $A \cdot b$, we are essentially projecting vector b onto the eigenvectors of matrix A .

And projection usually looks like this, $p = P \cdot b$ where P is the projection matrix.

So for the 2D case we can express this new information as,

$$\begin{aligned} A \cdot b &= (\lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T) b \\ &= \lambda_1 q_1 q_1^T b + \lambda_2 q_2 q_2^T b \\ &= P_1 b + P_2 b \end{aligned}$$

So the resultant vector $A \cdot b$ is the sum of 1D projection vectors of b onto the eigenvectors of A .

[1D because $q_1 q_1^T$ is rank 1]

This result is important for PCA later on.

Positive Semi-Definite Matrices (PSD)

A matrix is positive semi-definite if it is symmetric and all eigenvalues are non-negative.

Properties:

A matrix M is positive semi-definite if $x^T M x \geq 0$ for all $x \neq 0$

Consider the case where x is an eigenvector,

$$x^T M x = x^T \lambda x = \lambda x^T x$$

For a PSD matrix, all eigenvalues ≥ 0 and $x^T x \geq 0$.

Therefore, for all eigenvectors x ,

$$x^T M x \geq 0 \text{ for all } x \neq 0$$

The proof is complicated but it can be shown that this also holds for any vector x and not just eigenvectors.

Intuition for this, **PSD matrices are similar to positive numbers**, when you multiply vector x by a PSD matrix M , then it will never change the "sign" like how multiplying x by any positive number will still be a positive number.

Taking this intuition further,

We can write a real positive number as the square of two real numbers.

Similarly, we can also state the following for PSD matrices,

$$\forall \text{ PSD Matrices } M, \exists \text{ matrix } B \text{ such that } M = B^T B$$

Proving this is relatively simpler,

[Note that x are eigenvectors and ofc orthonormal in this case because M is PSD]

$$M = \sum_i \lambda_i x_i x_i^T$$

Define $y_i = \sqrt{\lambda_i} x_i$

This definition is possible as λ_i is non-negative and real

$$M = \sum_i y_i y_i^T = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix} \begin{bmatrix} y_1^T \\ \dots \\ y_n^T \end{bmatrix} = B^T B$$

Define B^T as the matrix formed by column vectors y_i

Can also write this as,

$$B^T = Q \Lambda^{1/2}$$

This also will be useful in many applications.

Positive Definite Matrices

A matrix is positive definite if it is symmetric and all eigenvalues are positive (eigenvalues cannot be zero).

So PD matrices are a subset of PSD matrices.

Therefore, $x^T M x > 0$ for all $x \neq 0$

Question: If A and B are PSD Matrices, then would A + B also be PSD?

Answer is Yes!

Remember the intuition that PSD are just positive numbers for matrices.

But if we need something concrete,

$$\begin{aligned} A + B &= x^T (A + B) x \\ &= x^T A x + x^T B x \\ &= \geq 0 \ \& \ \geq 0 \end{aligned}$$

$$A + B \geq 0$$

Same logic also holds for PD matrices.

Power Iteration Method

To Find Maximum Eigenvalue/Eigenvector

To get the Eigenvalues we need to solve C.E. equation which can be of really high degrees for large matrices.

And sometimes we don't even need all of the eigenvectors, only the dominant (maximum) eigenvalue is sometimes all we need.

Claim: If A is diagonalizable and has a dominant eigenvalue (largest eigenvalue is much larger than the second largest eigenvalue) then the power iteration sequence, $Ax, A^2x, A^3x, \dots$ converges to the dominant eigenvector (scaled).

Diagonalizable $\Rightarrow A$ can be written as, $A = S\Lambda S^{-1}$

Since A is diagonalizable, it has n linearly independent eigenvectors $v_1, \dots, v_n$ and the corresponding eigenvalues are $\lambda_1, \dots, \lambda_n$

Let v_1 and λ_1 be dominant.

Any vector x_0 in n -dimensional space can be represented as,
 $x_0 = c_1v_1 + \dots + c_nv_n$

Multiply this by A ,

$$Ax_0 = c_1Av_1 + \dots + c_nAv_n = c_1\lambda_1v_1 + \dots + c_n\lambda_nv_n$$

Again multiply this by A ,

$$A \cdot Ax_0 = c_1\lambda_1Av_1 + \dots + c_n\lambda_nAv_n = c_1\lambda_1^2v_1 + \dots + c_n\lambda_n^2v_n$$

So we can get,

$$A^kx_0 = c_1\lambda_1^kv_1 + \dots + c_n\lambda_n^kv_n$$

$$A^kx_0 = \lambda_1^k \left(c_1v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k + \dots + c_n \left(\frac{\lambda_n}{\lambda_1} \right)^k v_n \right)$$

Now we assumed λ_1 to be dominant, so the ratio of the eigenvalues with λ_1 would end up becoming really small the more iterations we do this for, i.e. the higher the value of k is. So for large enough k , the ratio of the eigenvalues would be really close to zero.

Since λ_1 is dominating, the ratios,

$$\left(\frac{\lambda_i}{\lambda_1} \right)^k \rightarrow 0 \text{ as } k \rightarrow \infty \text{ for all } i$$

Now,

$$A^kx_0 = \lambda_1^k c_1 v_1 = v'_1$$

λ_1 and c_1 are scalars, so we can assume that the output vector is a scaled version of the eigenvector v_1 .

Divide v'_1 by its norm to get the unit vector in the same direction.

19/10/21

Dominant Eigenvector Application: Page Rank Algorithm

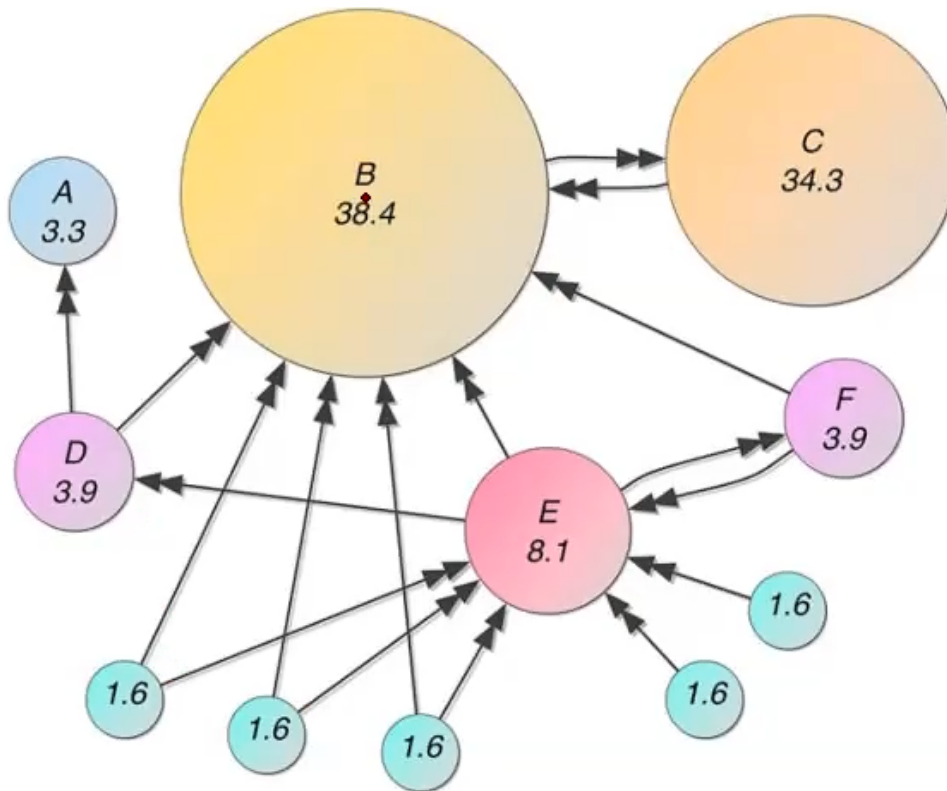
(by Google)

Importance of a page is indicated by its page rank.

The more incoming links a page has, the more important it becomes.

Outgoing links from an important page will have more weightage in determining importance for the linked page.

A page has higher rank if it is linked by other pages with high page ranks.



Page Rank of Web pages

Look at how important C becomes even though it has just one incoming link from B (but B is very important).

The definition of page rank includes page rank, so you cannot determine page rank for a single page, instead you must do it for the entire web/graph.

$L_{ij} = 1$ if webpage j links to webpage i (written $j \rightarrow i$)

$m_j = \sum_{k=1}^n L_{kj} \implies$ Total number of webpages that j links to

$$L = \begin{bmatrix} L_{11} & \dots & L_{1n} \\ \dots & \dots & \dots \\ L_{n1} & \dots & L_{nn} \end{bmatrix}$$

Note that for computing m_j we are summing the entries in the j^{th} column of L .

Page Rank of Page i is given as,

$$p_i = \sum_{j \rightarrow i} \frac{p_j}{m_j}$$

This just follows from the theory discussed above.

To determine the page rank of i , we look at all nodes j which have an outgoing edge to i .

m_j will tell you total links that j has and p_j will tell you page rank of j , i.e. how important node j is.

So we are considering both, the importance of the page as well as how many outgoing links that page has to determine the weightage it has in determining the page rank for node i .

Another thing that we are considering is that if page rank is more and number of outgoing links are less, then each outgoing link (if any) will have significantly higher weightage as compared to other node links.

For example, check the $B \rightarrow C$ edge in the graph, B has a high rank and only 1 outgoing edge, so that edge has a lot of weight in determining importance for C .

For each node i , we look at all nodes that are connected to it and this can be expressed in terms of matrix L as looking at the i^{th} row and only considering those j 's which have $L_{ij} = 1$

So we can rewrite p_i equation as,

$$p_i = \sum_{j=1}^n L_{ij} \cdot \frac{p_j}{m_j}$$

Do not worry about the initial value of p right now.

We also define the **Importance Matrix**,

$$A = \begin{bmatrix} \frac{L_{11}}{m_1} & \dots & \frac{L_{1n}}{m_n} \\ \dots & \dots & \dots \\ \frac{L_{n1}}{m_1} & \dots & \frac{L_{nn}}{m_n} \end{bmatrix}$$

Note that the sum of each column in A is 1.

Because $L_{11} + \dots + L_{n1} =$ Number of outgoing links that j has $= m_1$

So the numerator and denominator of the column sum are the same and thus 1.

Same point repeats for all the other columns.

Now our goal is to find the vector p which contains the final page ranks for each of the n nodes.

$$p = \begin{bmatrix} p_1 \\ \dots \\ p_n \end{bmatrix}$$

Now note that,

$$A \cdot p = \begin{bmatrix} \frac{L_{11}}{m_1} & \dots & \frac{L_{1n}}{m_n} \\ \dots & \dots & \dots \\ \frac{L_{n1}}{m_1} & \dots & \frac{L_{nn}}{m_n} \end{bmatrix} \cdot \begin{bmatrix} p_1 \\ \dots \\ p_n \end{bmatrix} = \begin{bmatrix} p_1 \\ \dots \\ p_n \end{bmatrix}$$

Therefore we get to this equation,

$$A \cdot p = p \implies p \text{ is eigenvector of } A \text{ with } \lambda = 1$$

Thus we have one equation and only one unknown 'p', so we can solve to find it.

The key thing to note now is that in a matrix as long as all the columns sum up to 1 and all the entries in the matrix are positive, the matrix has a dominant eigenvalue and it is 1!

[Proof for this is out of scope ^_^]

Such matrices are called **column stochastic matrices**.

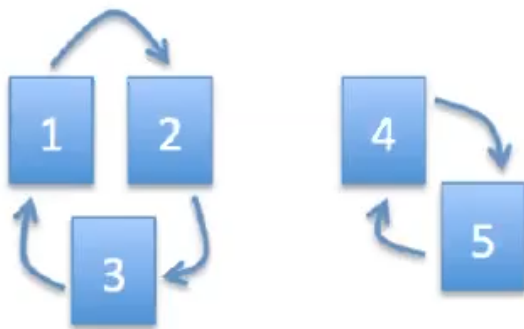
This means that A has dominant eigenvalue = 1.

So p is the eigenvector associated with the dominant eigenvalue 1 and we can use the power iteration method for it.

In practice Google implemented this a bit differently in their original paper.

We get the links in the matrix L and consequently A by analysing user data that is obtained. However in this application, even if there are no discernible links between two pages from user data, a random chance is allotted to indicate that any page can link to any page. This is represented as the Google Matrix.

Consider the following example,



$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

The Google matrix for this is given as,

$$A = \frac{0.15}{5} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} + 0.85 \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

What's happening here?

85% of the time, a web page will link to the web pages that it links to using the user data (original matrix A).

However 15% of the time, a web page can link to any other webpage.

We divide 0.15 by 5 because there are 5 pages, so the probability gets divided up in 5 equal portions.

$$A = \begin{bmatrix} 0.03 & 0.03 & 0.88 & 0.03 & 0.03 \\ 0.88 & 0.03 & 0.03 & 0.03 & 0.03 \\ 0.03 & 0.88 & 0.03 & 0.03 & 0.03 \\ 0.03 & 0.03 & 0.03 & 0.03 & 0.88 \\ 0.03 & 0.03 & 0.03 & 0.88 & 0.03 \end{bmatrix}$$

Note that this is also a column stochastic matrix, so the dominant eigenvalue is still 1 and we can find the dominant eigenvector.

Also note that in reality A will have millions and millions of rows and columns, but the matrix itself will be quite sparse as most web pages will be linked to only a few other web pages. Because of this, the dominant eigenvalue (and vector) will be very dominant and the next eigenvalue will be quite miniscule in comparison.

Now we can apply power iteration on this,

$$p^{(1)} = A \cdot p^{(0)}$$

$$p^{(2)} = A \cdot p^{(1)}$$

...

$$p^{(k)} = A \cdot p^{(k-1)}$$

Note:

$$p^{(0)} \implies \text{Random vector in the column space of } A$$

It plays the role of the x_0 vector that was described in the power iteration method.

Sir said that he read that power iteration converged in around 50 iterations only because of the high sparsity of the matrix.

Singular Value Decomposition

Problems with the general Eigen decomposition, $A = S \Lambda S^{-1}$,

- Doesn't work with rectangular matrices.
- S is usually not orthonormal (unless A is symmetric).

(Assume A is $m \times n$ throughout)

Goal of SVD:

Find orthonormal bases in the rowspace of A as well as in the column space of A such that A maps from rowspace basis to the column space basis.

In simpler words, we have to find two sets of basis vectors for matrix A , one is in its column space and the other is in its rowspace.

And these bases should be such that $A * v_i = u_i$ where v_i is a basis vector of the rowspace of A and u_i is a basis vector of the column space of A .

So the basis vector should map from R^n (row space) to R^m (column space).

Orthogonalization of Rowspace of A

Let r be the rank of A .

Select orthonormal basis $v_1, \dots, v_r$ in R^n such that it spans rowspace of A .

One way to do this is by using the Gram-Schmidt algorithm.

Continue the process to find $v_{r+1}, \dots, v_n$ in R^n such that it spans the nullspace of $A \Rightarrow$ This is done for completion of the process and doesn't play much of a role.

Now we also have to find basis vectors of column space of A .

By the mapping constraint that we have,

$$\text{for } i = 1, \dots, r, u_i = Av_i$$

All vectors $\{v_i\}$ are orthogonal by construction, but because of this constraint all $\{u_i\}$ are not necessarily orthogonal.

We want to find $\{v_i\}$ such that $\{u_i\}$ are also orthogonal.

So our new goal becomes to find the following,

$u_1, \dots, u_r =$ Orthonormal basis for column space of A

$u_{r+1}, \dots, u_n =$ Orthonormal basis for nullspace of A

$v_1, \dots, v_r =$ Orthonormal basis for rowspace of A

$v_{r+1}, \dots, v_m =$ Orthonormal basis for left nullspace of A

And we have to show that we can write A as,

$$A = U\Sigma V^T$$

$$U = \begin{bmatrix} u_1 & \dots & u_m \end{bmatrix}, V = \begin{bmatrix} v_1^T \\ \dots \\ v_n^T \end{bmatrix}, \Sigma = \text{Diagonal Matrix}$$

Orthogonal Basis of Rowspace using Eigenvectors

Let $\{v_i\}$ be eigenvectors of $A^T A$ with λ_i being the corresponding eigenvalues.

Because $A^T A$ is symmetric, its eigenvectors are orthogonal to each other.

We can use Eigendecomposition to write this as follows,

$$A^T A v_i = \lambda_i v_i \text{ \& } A = V \Lambda V^T$$

(where v_i are the columns of V)

Does this matrix V match the matrix V that we want to compute as part of our objective?

Check it...

$$A = U\Sigma V^T$$

$$\begin{aligned} A^T A &= (U \cdot \Sigma \cdot V^T)^T \cdot (U \cdot \Sigma \cdot V^T) \\ &= (V^T)^T \Sigma^T U^T \cdot U \cdot \Sigma \cdot V^T \end{aligned}$$

Our goal is to find U such that it is orthonormal, so we can assume $U^T U = I$

$$A^T A = V \cdot \Sigma^2 \cdot V^T \rightarrow [\Sigma \text{ is a diagonal matrix}]$$

This means that the Σ matrix that we are trying to find and the Λ matrix of A are related to each other as, $\Lambda = \Sigma^2$

Also note that V is definitely an orthogonal basis for the rowspace of A because it satisfied the equation.

Orthogonal Basis for Columnspace

Now again the vectors u_i belonging to matrix U will have to be of the form, $u_i = A * v_i$

Consider two such vectors, u_i and u_j , are they orthogonal?

We need to compute their dot product.

$$\begin{aligned} u_i^T u_j &= (A v_i)^T (A v_j) \\ &= v_i^T A^T A v_j \\ &= v_i^T \lambda_j v_j \\ &= \lambda_j v_i^T v_j \implies v_i \text{ \& } v_j \text{ are orthogonal} \end{aligned}$$

$$u_i^T u_j = 0$$

This means that U will be an orthogonal matrix and our condition that $u_i = A * v_i$ is also satisfied!

So you could use Gram-Schmidt to get orthogonal matrix V but it is not guaranteed that matrix U will satisfy our conditions.

This method of using eigenvectors of $A^T A$ however works perfectly.

Now to make u_i orthonormal, we have to find $\|A \cdot v_i\|$

$$\begin{aligned}\|A \cdot v_i\|^2 &= (A \cdot v_i)^T (A \cdot v_i) \\ &= v_i^T (A^T A v_i) \\ &= v_i^T \lambda_i v_i \\ &= \lambda_i (v_i^T v_i) \\ &= \lambda_i\end{aligned}$$

$$\text{Let } u_i = \frac{A v_i}{\|A v_i\|} = \frac{A v_i}{\sqrt{\lambda_i}}, \text{ for } i = 1, \dots, r$$

If $r < m$, we extend this basis for $\mathbb{R}^m$ as well (but again, won't need this).

We don't need those because the eigenvalues associated for all vectors that are greater than rank r is 0 (Spend some time on this after reading all of the SVD stuff).

SVD Construction for Matrix A

A is $m \times n$ real matrix

V is obtained from diagonal factorization of $A^T A = V \Lambda V^T$

Let $\sigma_i = \sqrt{\lambda_i}$, then $u_i = \frac{1}{\sigma_i} A v_i$ or $A v_i = \sigma_i u_i$

Put $\{v_1, \dots, v_r\}$ in columns of V, $\{u_1, \dots, u_r\}$ in columns of U and $\{\sigma_1, \dots, \sigma_r\}$ in the diagonals of Σ

And we'll have,

$$A V = U \Sigma \quad (\text{check validity using matrix-vector column interpretation haxx})$$

Thus the SVD is,

$$A = U \Sigma V^T \rightarrow [\cdot \cdot V^{-1} = V^T]$$

Square root of the eigenvalues (elements of the Sigma matrix) are the singular values of SVD. Vectors of U and V are called left singular and right singular vectors, respectively.

Note that:

$$A = \sum_i \sigma_i u_i v_i^T$$

So the decomposition can be written as a sum of Rank-1 Matrices.

Linear Transformation of x via Ax can be considered as,

1. Linear transformation using individual matrices, $\sigma_i u_i v_i^T$

2. Summation of linear transformation using rank-1 matrices, $\sigma_i u_i v_i^T$

Linear Transformation of x via Ax can be considered in 3 stages as,

1. Input transformation of vector x with matrix $V^T \Rightarrow$ Project x onto rowspace of A (V)
2. Scaling with Matrix Sigma (scaling the projections up by their singular values).
3. Output vector formation with columns of Matrix U $\Rightarrow$ Project scaled projections of x onto column space of A (U).

SVD is unique to a certain extent (for now just think that it's unique).